

Leader selection based Multi-Objective Flow Direction Algorithm (MOFDA): A novel approach for engineering design problems

Nima Khodadadi^a, Mohammad Ehteram^b, Hojat Karami^{b,*}, Mohammad H. Nadimi-Shahraki^c, Laith Abualigah^d, Seyedali Mirjalili^{e,f,g}

^a Department of Civil and Architectural Engineering, University of Miami, Coral Gables, FL, USA

^b Department of Water Engineering and Hydraulic Structures, Faculty of Civil Engineering, Semnan University, Semnan, Iran

^c Faculty of Computer Engineering, Najafabad Branch, Islamic Azad University, Najafabad 8514143131, Iran

^d Computer Science Department, Al al-Bayt University, Mafrq 25113, Jordan

^e Centre for Artificial Intelligence Research and Optimisation, Torrens University, Australia

^f University Research and Innovation Center (EKIK), Obuda University, Budapest, 1034, Hungary

^g Faculty of Electrical Engineering and Computer Science, VSB-TU Ostrava, Czech Republic, Ostrava, 70080, Czech Republic

ARTICLE INFO

Keywords:

Multi-objective optimization
Flow Direction Algorithm
Pareto optimality

ABSTRACT

Addressing complex real-world issues with conflicting objectives is a significant challenge in optimization. Practical algorithms must balance these objectives, mainly when decision-maker preferences are unclear. This paper introduces a multi-objective adaptation of the Flow Direction Algorithm (FDA) to address the shortcomings of traditional evolutionary and meta-heuristic optimization methods in multi-objective optimization (MOO). These conventional methods often fail to find Pareto optimal solutions and to represent all objectives fairly. Building on the FDA's success in single-objective tasks, we expanded its application to MOO, creating the Multi-Objective Flow Direction Algorithm (MOFDA). MOFDA incorporates new mechanisms to accurately and uniformly find optimal solutions for MOO challenges. It features a fixed-size external archive to maintain Pareto optimal solutions, uses a grid mechanism to improve non-dominated solutions within this archive, and implements a leader selection process to guide searches in the multi-objective space. These strategies enable MOFDA to discover superior solutions and ensure extensive coverage of the Pareto front. We validated MOFDA's effectiveness by testing it against 27 diverse problems using seven performance metrics. The results show MOFDA's ability to outperform well-known algorithms, achieving significant convergence and broad coverage, thus demonstrating its advanced capability in multi-objective optimization. The MOFDA source code is available at: <https://nimakhodadadi.com/algorithms-%2B-codes>.

1. Introduction

Advances in technology and computing systems are necessary for the development of human societies. Using new technologies to solve various problems and developing urban infrastructure are always key challenges for decision-makers [1]. Researchers and scientists in multiple fields have always tried to create computational systems based on programming and accurate calculations to solve various problems. Today, programming based on computer systems is one of the essential human solutions to solve multiple problems and develop human societies. Computer systems based on precision programming can increase computational accuracy and reduce computational time. Designers of

various techniques can reduce the error rate in calculations and improve the efficiency of various industrial projects by using accurate coding and designing advanced computing systems [2].

Urban development systems and their infrastructures require careful design, in which programmers can help human societies by using computer systems. One of the essential techniques in computer design and computational methods is optimization algorithms, which are widely used today due to accurate programming to solve various challenges [3]. Metaheuristic algorithms are utilized as search tools for finding the best optimal solutions in dealing with complex problems, in which upper-level techniques are used for dealing with complex issues. Red Deer Algorithm (RDA) [4], Tunicate Swarm Algorithm (TSA) [5],

* Corresponding author.

E-mail addresses: nima.khodadadi@miami.edu (N. Khodadadi), eh.mohammad@yahoo.com (M. Ehteram), hkarami@semnan.ac.ir (H. Karami), nadimi_m@yahoo.com (M.H. Nadimi-Shahraki), aligah.2020@gmail.com (L. Abualigah), ali.mirjalili@gmail.com (S. Mirjalili).

<https://doi.org/10.1016/j.rineng.2024.103670>

Received 20 September 2024; Received in revised form 19 November 2024; Accepted 5 December 2024

Available online 5 December 2024

2590-1230/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Greylag Goose Optimization (GGO) [6], Pathfinder Algorithm (PA) [7], Stochastic Paint Optimizer (SPO) [8], Golden Eagle Optimizer (GEO) [9], Electric Eel Foraging Optimization (EEFO) [10], Gradient-Based Optimizer (GBO) [11], Butterfly Optimization Algorithm (BOA) [12], Synergistic Swarm Optimization Algorithm (SSOA) [13], Sailfish Optimizer (SO) [14] and Hippopotamus Optimization (HO) [15] are some latest state-of-art algorithms.

In addition, some enhanced [16–18], binary [19–21], and hybridized [22–24] metaheuristics have been proposed for various applications. Inspired by the life of creatures in nature or based on the laws of physics and mathematics, optimization algorithms try to solve a specific problem using the optimization process [25]. To solve various problems and challenges with optimization algorithms, first, the problem is defined as a mathematical problem. Then, the algorithm seeks to provide the optimal answer based on the optimization process. However, solving a problem based on optimization algorithms faces various challenges, such as conflicting goals, local optimization, convergence speed, and finding the absolute optimal solution [26]. One of the complex problems that optimization algorithms face is multi-objective problems.

Multi-objective problems have multiple objective functions with more than one answer instead of single-objective optimization problems [27]. The main challenge is finding the answer in a situation where the target functions may conflict. The literature on multi-objective optimization using stochastic optimization methods primarily distinguishes between two approaches: a posteriori and a priori. In a priori methods, the multi-objective optimization problem is transformed into a single-objective issue by combining the objectives into one, using a set of weights to signify the importance of each objective, typically determined by an expert in the field. This aggregation allows the problem to be solved using single-objective optimization algorithms. However, these approaches have significant limitations, including the necessity to run the algorithm multiple times to identify the Pareto optimal set and the need for expert input. Additionally, certain specific Pareto optimal fronts may need to be more attainable through these methods [28].

On the other hand, posterior methods retain the multi-objective nature of the problem, aiming to find the Pareto optimal set in just one run. These approaches can identify any Pareto front, offering a broad applicability. Despite their advantages, such methods come with the drawbacks of higher computational demands and the complexity of managing multiple objectives simultaneously. According to the literature, a posteriori method has been extensively adopted since its inception, effectively addressing real-world challenges [29]. The initiation of a new multi-objective algorithm can be motivated by several elements.

The superior efficiency of this multi-objective algorithm surpasses existing ones, possibly due to the incorporation of novel techniques, optimization procedures, and design practices. The increased efficiency has the potential to make the algorithm more suitable for real-world implementations. The discovery of a currently unexplored algorithm has the potential to provide novel insights into the field of multi-objective optimization. Through the performance of a novel approach, it is feasible to contribute to advancing the field and initiate revolutionary resolutions to complicated problems. The challenges of multi-objective optimization are diverse and may require customized algorithms customized to their specific requirements. By introducing a new algorithm, its advantages can be evaluated in comparison to state-of-the-art algorithms, highlighting its superiority. Those above could enhance the proof of the effectiveness of the novel algorithm and establish a benchmark for future tests.

Karami et al. [30] recently introduced a new optimization algorithm called the flow direction algorithm (FDA). The FDA algorithm is based on the movement of water flow in watersheds. Examining the FDA's performance on several mathematical functions and complex engineering problems showed that the FDA had greater accuracy and convergence speed than other evolutionary algorithms. Plus, the FDA is easy to implement. The FDA, based on advanced operators, can also exit local optimizations. Therefore, the high capability of the FDA over other

evolutionary algorithms causes the present study to address how the FDA single-objective algorithm can be converted to the FDA multi-objective algorithm and then compare the performance of the FDA multi-objective algorithm on problems with multiple objective functions with the performance of other evolutionary algorithms.

The purpose of mentioning urban development challenges above is to highlight the relevance and applicability of multi-objective optimization methods in addressing complex real-world problems. Urban development projects often involve allocating resources across various objectives, such as minimizing costs, maximizing accessibility, and reducing environmental impact. Multi-objective optimization algorithms can assist urban planners in finding optimal resource allocation strategies by simultaneously considering multiple conflicting objectives. For example, it can help decide the optimal locations for new infrastructure projects while considering cost-effectiveness, environmental sustainability, and social equity. Urban development involves making decisions about land use, including zoning regulations, residential and commercial developments, and green spaces. These algorithms can support urban planners in optimizing land use plans by balancing competing objectives such as economic growth, environmental conservation, and social equity.

The No-Free Lunch (NFL) [31] theorem for optimization underscores the rationale for developing new or refining existing algorithms, demonstrating that no single optimization algorithm can address every problem. This principle applies to both single-objective and multi-objective optimization strategies. This study introduces a multi-objective version of the recently developed FDA to address complex optimization challenges involving multiple objectives. While existing algorithms can tackle a broad spectrum of issues, the NFL theorem clarifies their limitations in providing universal solutions. Consequently, this research presents the multi-objective FDA with the aspiration of more effectively solving specific existing or novel problems. To summarize, this work presents significant contributions as follows:

- The MOFDA algorithm was formulated as a multi-objective variant of the FDA algorithm, employing its inherent capabilities.
- The MOFDA integrates a fixed-sized external archive to capture and generate Pareto optimal solutions.
- For the creating MOFDA, an archive mechanism was used to store and retain the derived Pareto optimal solutions.
- A grid mechanism was used in MOFDA to enhance non-dominated solutions in the archive.
- The efficacy of the method above was evaluated by comparing it with three contemporary multi-objective optimization (MOO) algorithms on a set of 27 mathematical, engineering, and truss structure problems.
- Statistical analysis was conducted to differentiate the MOFDA algorithm from other selected algorithms.
- The application of MOFDA in enhancing engineering and truss structure problems highlights its potential to address practical challenges.
- The article additionally presented visual representations of the Pareto sets and Pareto fronts generated by the MOFDA algorithm.
- The MOFDA algorithm was superior in both qualitative and quantitative analyses.
- The MOFDA was compared to Multi-Objective Particle Swarm Optimization (MOPSO) [32], Multi-Objective Grey Wolf Optimizer (MOGWO) [33], Multi-Objective Multi-Verser Optimizer (MOMVO) [34], Two-archive Multi-objective Multi-verser Optimizer (MOMVO2arc) [35], Multi-objective Cheetah Optimizer (MOCO) [36] and Multi-Objective Hippopotamus Optimizer (MOHO) [37].
- Seven metrics, Generational Distance (GD), Spacing (S), Maximum Spread (MS), Inverted Generational Distance (IGD), Hypervolume (HV), Space to extend (STE), and the Wilcoxon Rank-Sum Test

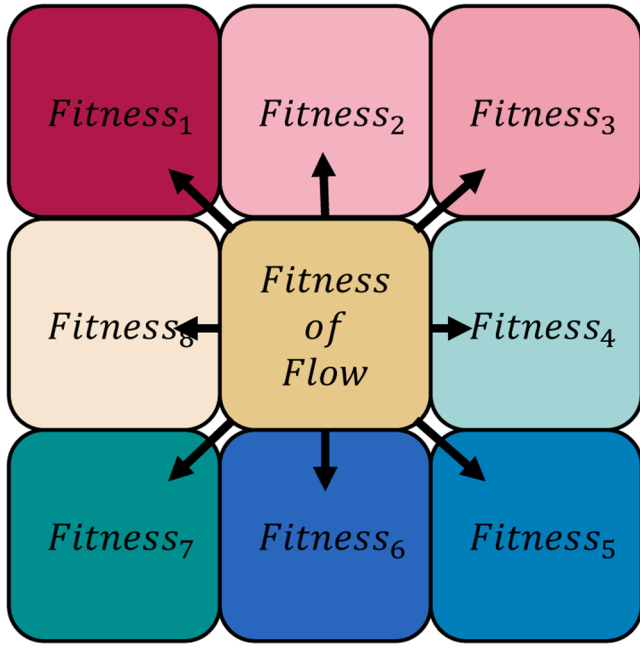


Fig. 1. Flow and eight locations around it.

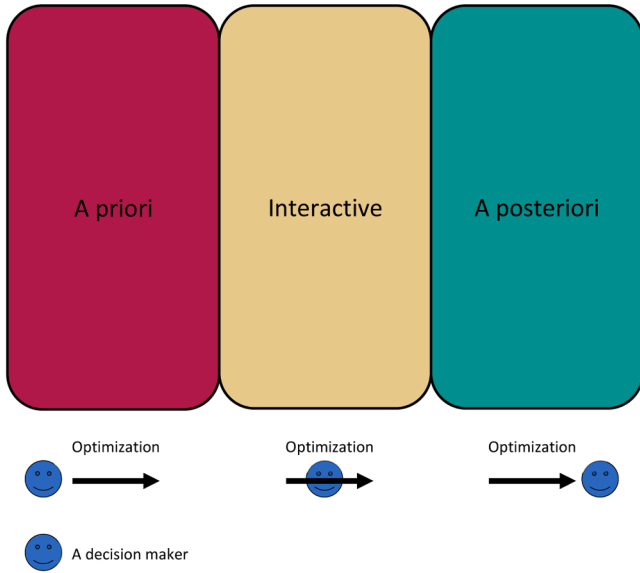


Fig. 2. Three categories of multi-objective optimization based on decision-making

Table 1
Parameters setting of all algorithms.

Parameters	MOPSO	MOGWO	MOMVO	MOFDA
Rate of mutation	0.5	N/A	N/A	N/A
The number of solutions in each population	100	100	100	100
The number of solutions in the archive	100	100	100	100
The number of archive grids for each objective	30	30	30	30
C ₁ C ₁	1	N/A	N/A	N/A
C ₂ C ₂	2	N/A	N/A	N/A
W	0.4	N/A	N/A	N/A
β	4	4	4	4
γ	2	2	2	2

(WRST), were employed to demonstrate MOFDA's robustness and superior performance.

- One advantage of the current study was that different functions and problems were used to assess the performance of other optimizers.

The rest of the paper is organized as follows: Section 2 provides an in-depth literature review of multi-objective optimization. Section 3 offers the original single-objective Flow Direction Algorithm and its proposed multi-objective variant. Section 4 provides results and discussions. Finally, Section 5 gives conclusions and future remarks.

2. Related works

Solving single-objective optimization problems is more straightforward than multi-objective optimization problems due to only one unique answer and one objective function. Comparing the answers and choosing the best absolute optimal answer in single-objective optimization problems is more accessible than in multi-objective. Multi-objective optimization problems have more than one solution and are defined based on the following mathematical format [29]:

$$F(\vec{z}) = \{f_1(\vec{z}), f_2(\vec{z}), \dots, f_o(\vec{z})\} \quad (1)$$

$$\text{subject}(to) : y_i(\vec{z}) \geq 0, i = 1, 2, \dots, m \quad (2)$$

$$v_i(\vec{z}) = 0, i = 1, 2, \dots, p \quad (3)$$

$$Lo_i \leq z_i \leq Up_i \quad (4)$$

Where O : the number of objective functions, p : the number of equality constraint, where n : the number of variables, y_i : the inequality constraint, v_i : the equality constraint, Lo_i : the lower bound of the decision variable, and Up_i : the upper bound of the decision variable.

To define a multi-objective optimization problem, consider the two vectors $\vec{z} = (z_1, z_2, \dots, z_k)$ and $\vec{c} = (c_1, c_2, \dots, c_k)$. The vector $\vec{z}$ is said to denote $\vec{c}$ ($\vec{z} \prec \vec{c}$) if and only if:

$$\forall i \in \{1, 2, \dots, k\} : f_i(\vec{z}) \leq f_i(\vec{c}) \wedge \exists i \in \{1, 2, \dots, k\} : f_i(\vec{z}) < f_i(\vec{c}) \quad (5)$$

Pareto optimality: a solution $\vec{z} \in Z$ is considered as Pareto optimality if and only if:

$$P_a = \{\vec{z} \in Z \mid \nexists \vec{c} \in Z : \vec{c} \prec \vec{z}\} \quad (6)$$

Pareto optimal set: The set of all Pareto optimal solutions is called the Pareto set, which is defined based on the following equation:

$$P_a = \{\vec{z}, \vec{c} \in Z \mid \nexists \vec{z} \in Z : \vec{c} \prec \vec{z}\} \quad (7)$$

Pareto Front: A set contains the values of the objective function for the set of Pareto solutions, which is defined based on the following equation:

$$P_f = \{f(\vec{z}) \mid \vec{z} \in P_s\} \quad (8)$$

Solving a multi-objective problem requires finding the optimal Pareto set. Various multi-objective algorithms have been proposed to solve such problems [38,39].

Abbas et al. [40] used evolutionary algorithms to solve multi-objective optimization problems. They introduced Pareto-frontier differential evolution (PDE) for solving multi-objective optimization problems. The PDE was tested on standard test problems. They concluded that the PDE outperformed the Strength Pareto Evolutionary Algorithm (SPEA). Li et al. [41] proposed a modified particle swarm optimization (PSO), a non-dominated sorting PSO (NSPSO) method for solving multi-objective optimization problems. The NSPSO improved the use of particles' personal bests for multi-objective optimization problems. Their results indicated that the NSPSO outperformed the

Table 2
CEC-09 benchmark functions with two different objectives.

Function	Mathematical formulation	D	Range
UF1	$f_1 = x_1 + \frac{2}{ J_1 } \sum_{j \in J_1} \left[x_j - \sin \left(6\pi x_1 + \frac{j\pi}{n} \right) \right]^2,$ $f_2 = 1 - \sqrt{x_1} + \frac{2}{ J_2 } \sum_{j \in J_2} \left[x_j - \sin \left(6\pi x_1 + \frac{j\pi}{n} \right) \right]^2$ $J_1 = \{j j \text{ is odd and } 2 \leq j \leq n\}, \quad J_2 = \{j j \text{ is even and } 2 \leq j \leq n\}$	30	$x_1 \in [0,1]$ $x_i \in [-1,1]$ $i = 1, \dots, D$
UF2	$f_1 = x_1 + \frac{2}{ J_1 } \sum_{j \in J_1} y_j^2, \quad f_2 = 1 - \sqrt{x_1} + \frac{2}{ J_2 } \sum_{j \in J_2} y_j^2$ $J_1 = \{j j \text{ is odd and } 2 \leq j \leq n\}, \quad J_2 = \{j j \text{ is even and } 2 \leq j \leq n\}$ $x_j = \begin{cases} 0.3x_1^2 \cos(24\pi x_1 + \frac{4j\pi}{n}) + 0.6x_1 \cos(6\pi x_1 + \frac{j\pi}{n}) & \text{if } j \in J_1 \\ 0.3x_1^2 \cos(24\pi x_1 + \frac{4j\pi}{n}) + 0.6x_1 \cos(6\pi x_1 + \frac{j\pi}{n}) & \text{if } j \in J_2 \end{cases}$	30	$x_1 \in [0,1]$ $x_i \in [-1,1]$ $i = 1, \dots, D$
UF3	$f_1 = x_1 + \frac{2}{ J_1 } \left(4 \sum_{j \in J_1} y_j^2 - 2 \prod_{j \in J_1} \cos \left(\frac{20y_j\pi}{\sqrt{j}} \right) + 2 \right)$ $f_2 = \sqrt{x_1} + \frac{2}{ J_2 } \left(4 \sum_{j \in J_2} y_j^2 - 2 \prod_{j \in J_2} \cos \left(\frac{20y_j\pi}{\sqrt{j}} \right) + 2 \right)$ $J_1 \text{ and } J_2 \text{ are the same as those of UF1, } y_j = x_j - x_1^{0.5 \left(1.0 + \frac{3(j-2)}{\pi-2} \right)}, \quad j = 2, 3, \dots, n$	30	$x_i \in [0,1]$
UF4	$f_1 = x_1 + \frac{2}{ J_1 } \sum_{j \in J_1} h(y_j), \quad f_2 = 1 - x_2 + \frac{2}{ J_2 } \sum_{j \in J_2} h(y_j)$ $J_1 \text{ and } J_2 \text{ are the same as those of UF1, } y_j = x_j - \sin \left(6\pi x_1 + \frac{j\pi}{n} \right), \quad j = 2, 3, \dots, n$ $h(t) = \frac{ t }{1 + e^{2 t }}$	30	$x_1 \in [0,1]$ $x_i \in [-2,2]$ $i = 1, \dots, D$
UF5	$f_1 = x_1 + \left(\frac{1}{2N} + \varepsilon \right) \sin(2N\pi x_1) + \frac{2}{ J_1 } \sum_{j \in J_1} h(y_j)$ $f_2 = 1 - x_1 + \left(\frac{1}{2N} + \varepsilon \right) \sin(2N\pi x_1) + \frac{2}{ J_2 } \sum_{j \in J_2} h(y_j)$ $J_1 \text{ and } J_2 \text{ are identical to those of UF1, } \varepsilon > 0, y_j = x_j - \sin \left(6\pi x_1 + \frac{j\pi}{n} \right), \quad j = 2, 3, \dots, n$	30	$x_1 \in [0,1]$ $x_i \in [-1,1]$ $i = 1, \dots, D$
UF6	$h(t) = 2t^2 - \cos(4\pi t) + 1$ $f_1 = x_1 + \max \left\{ 0, 2 \left(\frac{1}{2N} + \varepsilon \right) \sin(2N\pi x_1) \right\} + \frac{2}{ J_1 } \left(4 \sum_{j \in J_1} y_j^2 - 2 \prod_{j \in J_1} \cos \left(\frac{20y_j\pi}{\sqrt{j}} \right) + 1 \right)$ $f_2 = 1 - x_1 + \max \left\{ 0, 2 \left(\frac{1}{2N} + \varepsilon \right) \sin(2N\pi x_1) \right\} + \frac{2}{ J_2 } \left(4 \sum_{j \in J_2} y_j^2 - 2 \prod_{j \in J_2} \cos \left(\frac{20y_j\pi}{\sqrt{j}} \right) + 1 \right)$ $J_1 \text{ and } J_2 \text{ are identical to those of UF1, } \varepsilon > 0, y_j = x_j - \sin \left(6\pi x_1 + \frac{j\pi}{n} \right), \quad j = 2, 3, \dots, n$	30	$x_1 \in [0,1]$ $x_i \in [-1,1]$ $i = 1, \dots, D$
UF7	$f_1 = \sqrt[3]{x_1} + \frac{2}{ J_1 } \sum_{j \in J_1} y_j^2, f_2 = 1 - \sqrt[3]{x_1} + \frac{2}{ J_2 } \sum_{j \in J_2} y_j^2$ $J_1 \text{ and } J_2 \text{ are identical to those of UF1, } \varepsilon > 0, y_j = x_j - \sin \left(6\pi x_1 + \frac{j\pi}{n} \right), \quad j = 2, 3, \dots, n$	30	$x_1 \in [0,1]$ $x_i \in [-1,1]$ $i = 1, \dots, D$

original version of the multi-objective PSO.

Zhang and Li [42] proposed a multi-objective evolutionary algorithm (MOEA). They used MEAO based on decomposition to solve different optimization problems. The MEAO received information from its several neighboring sub-problems to solve each sub-problem. The results indicated that the MEAO had lower computational complexity than the non-dominated sorting genetic algorithm II (NSGAI) [43]. Yang and Deb [44] introduced a new version of the cuckoo search (CS) algorithm to solve multi-objective optimization problems. They modified some assumptions of the original version of CA. They assumed the k th egg in the multi-objective CA corresponds to the solution to the k th objective.

Additionally, the cuckoos leave the nest with a probability of p_a . Afterward, a new nest was built with k eggs according to their similarities/ differences. The multi-objective CS outperformed the NSGAI and strength Pareto evolutionary algorithm (SPEA) [45].

Bhowmik and Chakraborty et al. [46] updated the gravitational search algorithm (GSA) for solving multi-objective optimization problems. The elitist, non-dominated sorting and crowding distance were used for the multi-objective GSA. They used an archive of the Pareto optimal solutions to provide some elitism. They reported the superiority of multi-objective GSA in solving complex multi-objective optimization problems. Mirjalili et al. [47] proposed a new multi-objective algorithm: a multi-objective antlion optimizer (MOALO). An archive was used to save non-dominated Pareto optimal solutions obtained so far. They used

MOALO to find the optimal solutions of unconstrained and constrained functions. The results indicated that the MOALO benefited from high convergence and coverage.

Mirjalili et al. [48] used a new multi-objective algorithm: a multi-objective grasshopper swarm optimization algorithm (MOGSOA). The grasshopper swarm algorithm acted based on the interaction of grasshoppers in the group, including attraction force, repulsion force, and comfort zone. They are used to archive and select targets to provide MOGSOA. The results indicated that the MOGSA provided high accuracy regarding obtained Pareto optimal solutions and their distributions. Dhiman et al. [29] proposed a multi-objective seagull optimization algorithm (MOSEOA) for solving problems with multiple objective functions.

A concept of the dynamic archive was used to save the non-dominated Pareto optimal solutions. They used roulette wheel selection to choose effective archiving solutions. They concluded that the MOSEOA was better than the other existing algorithms. The multi-objective Crystal Structure Algorithm (MOCryStAl) [49] is inspired by the principles underlying the formation of crystal structures. For the performance evaluation of this algorithm, some real-world engineering design problems are used to evaluate the efficiency of the proposed method. The results demonstrate that the proposed method can provide excellent results compared to other methods.

Table 3

CEC-09 benchmark functions with three different objectives.

Function	Mathematical formulation	D	Range
UF8	$f_1 = \cos(0.5x_1\pi)\cos(0.5x_2\pi) + \frac{2}{ J_1 } \sum_{j \in J_1} \left(x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right)$ $f_2 = \cos(0.5x_1\pi)\sin(0.5x_2\pi) + \frac{2}{ J_2 } \sum_{j \in J_2} \left(x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right)$ $f_3 = \sin(0.5x_1\pi) + \frac{2}{ J_3 } \sum_{j \in J_3} \left(x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right)$ $J_1 = \{j 3 \leq j \leq n, \text{and } j - 1 \text{ is a multiplication of } 3\}$ $J_2 = \{j 3 \leq j \leq n, \text{and } j - 2 \text{ is a multiplication of } 3\}$ $J_3 = \{j 3 \leq j \leq n, \text{and } j \text{ is a multiplication of } 3\}$	30	$x_1 \in [0,1]$ $x_2 \in [0,1]$ $x_i \in [-2,2]$ $i = 1, \dots, D$
UF9	$f_1 = 0.5[\max\{0, (1 + \varepsilon)(1 - 4(2x_1 - 1)^2)\}] +$ $\frac{2}{ J_1 } \sum_{j \in J_1} \left(x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right)$ $f_2 = 0.5[\max\{0, (1 + \varepsilon)(1 - 4(2x_1 - 1)^2)\} + 2x_1]x_2 +$ $\frac{2}{ J_2 } \sum_{j \in J_2} \left(x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right)$ $f_3 = 1 - x_2 + \frac{2}{ J_3 } \sum_{j \in J_3} \left(x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right)$ $J_1 = \{j 3 \leq j \leq n, \text{and } j - 1 \text{ is a multiplication of } 3\}$ $J_2 = \{j 3 \leq j \leq n, \text{and } j - 2 \text{ is a multiplication of } 3\}$ $J_3 = \{j 3 \leq j \leq n, \text{and } j \text{ is a multiplication of } 3\}, \quad \varepsilon = 0.1$	30	$x_1 \in [0,1]$ $x_2 \in [0,1]$ $x_i \in [-2,2]$ $i = 1, \dots, D$
UF10	$f_1 = \cos(0.5x_1\pi)\cos(0.5x_2\pi) + \frac{2}{ J_1 } \sum_{j \in J_1} [4y_j^2 - \cos(8\pi y_j) + 1]$ $f_2 = \cos(0.5x_1\pi)\sin(0.5x_2\pi) + \frac{2}{ J_2 } \sum_{j \in J_2} [4y_j^2 - \cos(8\pi y_j) + 1]$ $f_3 = \sin(0.5x_1\pi) + \frac{2}{ J_3 } \sum_{j \in J_3} [4y_j^2 - \cos(8\pi y_j) + 1]$ $J_1 = \{j 3 \leq j \leq n, \text{and } j - 1 \text{ is a multiplication of } 3\}$ $J_2 = \{j 3 \leq j \leq n, \text{and } j - 2 \text{ is a multiplication of } 3\}$ $J_3 = \{j 3 \leq j \leq n, \text{and } j \text{ is a multiplication of } 3\}$	30	$x_1 \in [0,1]$ $x_2 \in [0,1]$ $x_i \in [-2,2]$ $i = 1, \dots, D$

Table 4

Multimodal benchmark functions with fixed dimension.

Function	Mathematical formulation	D	Range
ZDT1	$F_1 = x_1, F_2 = g\left(1 - \sqrt{F_1/g}\right), g = 1 + \frac{9}{d-1} \sum_{i=2}^d x_i$	30	$x_i \in [0,1]$
ZDT2	$F_1 = x_1, F_2 = g\left(1 - (F_1/g)^2\right), g = 1 + \frac{9}{d-1} \sum_{i=2}^d x_i$	30	$x_i \in [0,1]$
ZDT3	$F_1 = x_1, F_2 = g\left(1 - \sqrt{F_1/g} - F_1/g \sin(10\pi F_1)\right), g = 1 + \frac{9}{d-1} \sum_{i=2}^d x_i$	30	$x_i \in [0,1]$
ZDT4	$F_1 = x_1, F_2 = g\left(1 - \sqrt{F_1/g}\right), g = 1 + 10(d-1) + \sum_{i=2}^d (x_i^2 - 10\cos(4\pi x_i))$	10	$x_1 \in [0,1]$ $x_i \in [-5,5]$ $i = 1, \dots, D$
ZDT6	$F_1 = 1 - \exp(-4x_1)\sin^6(6\pi x_1), F_2 = g\left(1 - (F_1/g)^2\right), g = 1 + 9\left(\frac{\sum_{i=2}^d x_i}{d-1}\right)^{0.25}$	10	$x_i \in [0,1]$
DTLZ2	$F_1 = (1 + g)\cos\left(x_1\left(\frac{\pi}{2}\right)\right)\cos\left(x_2\left(\frac{\pi}{2}\right)\right), F_2 = (1 + g)\cos\left(x_1\left(\frac{\pi}{2}\right)\right)\sin\left(x_2\left(\frac{\pi}{2}\right)\right), F_3 = (1 + g)\sin\left(x_1\left(\frac{\pi}{2}\right)\right), g = \sum_{i=3}^d (x_i - 0.5)^2$	12	$x_i \in [0,1]$
DTLZ4	$F_1 = (1 + g)\cos\left(x_1^\pi\left(\frac{\pi}{2}\right)\right)\cos\left(x_2^\pi\left(\frac{\pi}{2}\right)\right), F_2 = (1 + g)\cos\left(x_1^\pi\left(\frac{\pi}{2}\right)\right)\sin\left(x_2^\pi\left(\frac{\pi}{2}\right)\right), F_3 = (1 + g)\sin\left(x_1^\pi\left(\frac{\pi}{2}\right)\right), g = \sum_{i=3}^d (x_i - 0.5)^2$	12	$x_i \in [0,1]$

3. Single-objective and Multi-objective Flow Direction Algorithms

In this section, single-objective and multi-objective Flow Direction Algorithms are presented.

Table 5

Comparison of algorithms using the IGD performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
UF1	Ave	4.6991E-03	5.4723E-03	3.7062E-03	5.4488E-03
	SD	2.9090E-03	2.3573E-03	4.8802E-04	7.5051E-04
	WRST	+	+	-	-
UF2	Ave	4.2506E-03	3.1171E-03	2.9279E-03	2.6161E-03
	SD	2.8128E-04	1.4449E-04	4.5436E-04	3.2030E-04
	WRST	+	+	+	-
UF3	Ave	1.7374E-02	1.1859E-02	1.5356E-02	9.7440E-03
	SD	3.4505E-03	1.0876E-03	3.3450E-02	1.0248E-03
	WRST	+	+	+	-
UF4	Ave	2.7983E-03	2.7729E-03	3.2078E-03	2.2255E-03
	SD	2.8906E-04	4.2623E-04	3.7832E-04	2.7964E-04
	WRST	+	+	+	-
UF5	Ave	3.7349E-01	3.6605E-01	2.0045E-01	4.2706E-01
	SD	2.5436E-01	1.0235E-01	7.5647E-02	1.1429E-01
	WRST	+	+	-	-
UF6	Ave	2.7689E-02	1.6384E-02	1.6063E-02	1.5750E-02
	SD	7.1654E-03	1.4457E-03	7.0654E-03	1.1513E-02
	WRST	+	+	+	-
UF7	Ave	4.3657E-03	4.4653E-03	5.2978E-03	4.1821E-03
	SD	1.7843E-03	2.0537E-03	5.6578E-03	1.4452E-03
	WRST	+	+	+	-
UF8	Ave	8.8865E-03	4.2972E-03	3.9913E-03	3.9807E-03
	SD	2.3436E-03	1.3923E-03	3.4530E-03	1.2707E-03
	WRST	+	+	+	-
UF9	Ave	1.2243E-02	4.7086E-03	4.9919E-03	6.5907E-03
	SD	4.1453E-03	9.5194E-04	6.5467E-04	1.1975E-03
	WRST	+	-	+	-
UF10	Ave	6.0131E-02	3.6391E-02	1.2436E-02	2.2616E-02
	SD	3.4567E-02	9.0787E-03	6.5463E-03	4.4727E-03
	WRTWRST	+	+	-	-
W^+ / W^-		20/0	18/2	14/6	-
$+/-/=$		10/0/0	9/1/0	7/3/0	-

Table 6

Comparison of algorithms using the GD performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
UF1	Ave	2.4743E-02	3.4966E-02	3.5467E-03	3.0981E-02
	SD	2.6164E-02	3.1994E-02	3.5467E-02	1.3643E-02
	WRST	+	+	-	
UF2	Ave	1.3447E-02	1.1215E-02	4.4536E-02	1.0306E-02
	SD	3.5959E-03	2.1213E-03	3.4356E-03	5.3793E-03
	WRST	+	+	+	
UF3	Ave	4.1606E-02	7.2062E-02	6.5678E-02	2.6685E-02
	SD	6.6331E-02	1.0267E-02	2.5654E-02	4.3309E-03
	WRST	+	+	+	
UF4	Ave	3.4599E-02	1.0169E-02	2.4356E-02	7.1993E-03
	SD	1.3499E-02	3.4790E-03	1.0967E-03	9.1939E-04
	WRST	+	+	+	
UF5	Ave	3.2451E-01	2.8272E-01	2.7685E-01	2.0587E-01
	SD	2.3456E-01	1.5805E-01	7.5436E-02	1.2072E-01
	WRST	+	+	+	
UF6	Ave	8.7439E-02	1.1532E-01	5.3657E-02	1.9250E-01
	SD	4.8820E-02	5.0468E-02	2.5456E-01	1.0909E-02
	WRST	+	+	-	
UF7	Ave	6.8273E-03	9.1316E-03	5.5467E-03	2.8312E-02
	SD	3.5460E-03	1.2322E-02	3.0007E-03	1.3736E-02
	WRST	+	+	-	
UF8	Ave	1.1689E-01	3.3914E-02	6.6879E-02	1.2922E-01
	SD	4.4989E-01	1.3307E-02	6.1768E-01	1.9873E-01
	WRST	+	-	+	
UF9	Ave	4.0922E-01	5.1960E-02	6.5436E-02	4.9551E-01
	SD	8.2642E-02	2.1763E-02	5.1567E-02	1.3557E-01
	WRST	+	-	+	
UF10	Ave	5.9089E-01	6.9388E-01	2.0034E-01	1.8992E-01
	SD	1.9205E-01	2.0777E-01	4.3465E-01	1.4206E-01
	WRST	+	+	+	
W^+ / W^-		20/0	16/4	14/6	
$+/-/=$		10/0/0	8/2/0	7/3/0	

Table 8

Comparison of algorithms using the S performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
UF1	Ave	3.4569E-02	5.8711E-02	5.5608E-03	1.8854E-02
	SD	9.9871E-03	4.3083E-02	2.3200E-03	6.6951E-03
	WRST	+	+	-	
UF2	Ave	2.3490E-02	1.4426E-02	9.4563E-03	1.2132E-03
	SD	4.0090E-03	4.0894E-03	4.4590E-03	3.4930E-03
	WRST	+	+	+	
UF3	Ave	3.0845E-02	7.3997E-02	6.6579E-02	1.2055E-02
	SD	4.4540E-02	2.5622E-02	9.6579E-02	1.8554E-03
	WRST	+	+	+	
UF4	Ave	1.4350E-02	2.0138E-02	3.4309E-03	1.0022E-03
	SD	2.4563E-03	1.0792E-02	6.6657E-03	1.7367E-03
	WRST	+	+	+	
UF5	Ave	3.4590E-02	8.9494E-02	7.4356E-03	7.9232E-02
	SD	4.6579E-02	7.0701E-02	7.5690E-03	3.8513E-02
	WRST	+	+	-	
UF6	Ave	3.0098E-02	1.3105E-01	4.2010E-03	5.9846E-02
	SD	5.3345E-02	1.8372E-01	3.0082E-03	3.4918E-02
	WRST	+	+	-	
UF7	Ave	1.4360E-02	2.6653E-02	5.9087E-03	1.7508E-02
	SD	6.8790E-03	4.5707E-02	4.4350E-03	3.0398E-03
	WRST	+	+	-	
UF8	Ave	3.7887E-01	4.4312E-01	3.8780E-01	1.1341E-01
	SD	9.0098E-02	4.1641E-02	5.6579E-02	3.0444E-02
	WRST	+	+	+	
UF9	Ave	4.0088E-01	6.6954E-01	3.5096E-01	3.3279E-01
	SD	9.7877E-02	4.2886E-02	4.5467E-02	3.1706E-02
	WRST	+	+	+	
UF10	Ave	1.2134E+00	4.1590E-01	2.3452E-01	1.1652E-01
	SD	4.3244E-01	1.3872E-01	7.6578E-02	5.5509E-02
	WRST	+	+	+	
W^+ / W^-		20/0	20/0	12/8	
$+/-/=$		10/0/0	10/0/0	6/4/0	

Table 7

Comparison of algorithms using the MS performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
UF1	Ave	7.7689E-01	9.6206E-01	5.6342E-01	1.2897E+00
	SD	2.3489E-01	1.1721E-01	2.2353E-01	1.0336E-01
	WRST	+	+	+	
UF2	Ave	1.0064E+00	1.2007E+00	8.6758E-01	1.2117E+00
	SD	2.4569E-01	5.7151E-02	2.3908E-01	1.6684E-01
	WRST	+	+	+	
UF3	Ave	2.2027E+00	1.0923E+00	2.3400E+00	1.0000E+00
	SD	6.6524E-01	9.2630E-02	7.1675E-01	0.0000E+00
	WRST	+	+	-	
UF4	Ave	8.9867E-01	1.0049E+00	9.9356E-01	1.0336E+00
	SD	3.348E-02	8.6663E-03	2.0002E-02	8.2737E-03
	WRST	+	+	+	
UF5	Ave	7.5767E-01	1.2035E+00	9.0125E-01	2.7896E+00
	SD	8.6578E-01	8.6410E-01	9.6179E-01	7.0311E-01
	WRST	+	+	+	
UF6	Ave	7.6579E-01	1.0298E+00	5.5460E-01	2.7203E+00
	SD	9.4583E-01	3.7343E-02	5.3436E-01	1.0789E+00
	WRST	+	+	+	
UF7	Ave	9.8843E-01	1.0202E+00	6.6579E-01	1.4812E+00
	SD	3.4567E-01	4.9259E-02	4.4657E-01	2.5897E-01
	WRST	+	+	+	
UF8	Ave	4.9802E+00	9.3689E-01	1.0980E+00	2.8760E+00
	SD	1.4655E+00	2.1879E-01	9.9244E-01	2.7675E+00
	WRST	-	+	+	
UF9	Ave	6.5467E+00	1.4259E+00	2.5679E+00	8.4688E+00
	SD	4.6546E-01	4.8994E+00	9.6579E-01	2.1931E-01
	WRST	+	+	+	
UF10	Ave	7.6578E+00	6.3441E+00	1.3456E+00	3.8168E+00
	SD	3.4567E+00	1.4704E+00	2.2348E+00	2.1496E+00
	WRST	-	+	+	
W^+ / W^-		16/4	20/0	18/2	
$+/-/=$		8/2/0	10/0/0	9/1/0	

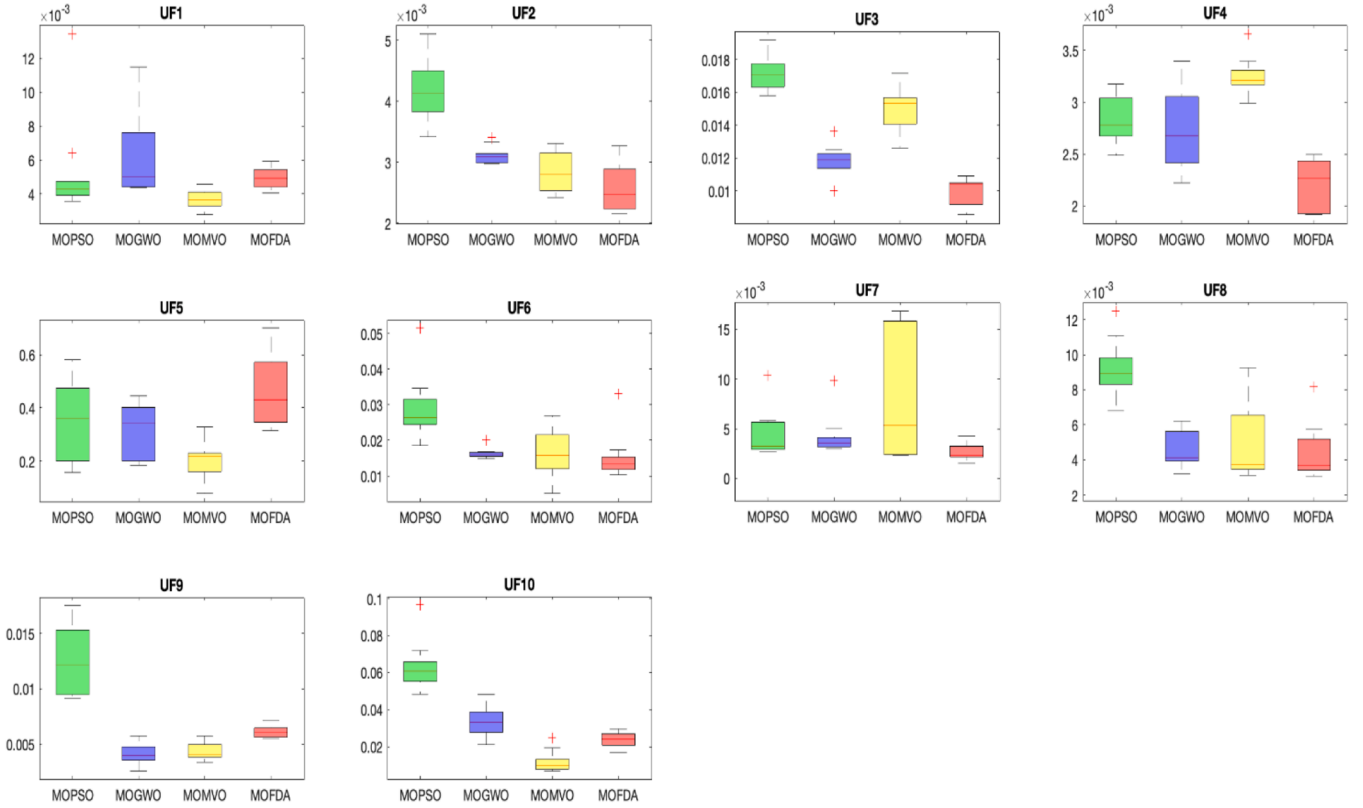


Fig. 3. Boxplot of the statistical results (IGD) for CEC-09 problems.

3.1. Single-objective Flow Direction Algorithm

The amount of adequate rainfall in catchments is the amount of rainfall that flows on the surface but does not penetrate the ground. Adequate rainfall is the amount of water remaining on the earth's surface after loss processes such as evapotranspiration and infiltration, known as runoff. One method of calculating the amount of runoff is using the ϕ index (cm/hr), which shows the average loss during rainfall. The amount of rainfall is obtained from the difference between rainfall and ϕ index in each step.

$$ra_d = \sum_{m=1}^M (Re_m - \phi \Delta t) \quad (9)$$

Where ra_d : the amount of runoff, Re_m : the amount of rainfall, Δt : the time interval, and ϕ : the average amount of water loss

The runoff moves towards the basin outlet in terms of the slope. To simulate the process of runoff movement, the basin is divided into different cells. The amount of runoff in each cell is transferred to the adjacent cell based on the slope and altitude of the neighboring cells. This method assumes that each cell has eight neighbors with height and distance from adjacent cells (Fig. 1). The difference in size and distance between each cell and the adjacent cell determines the flow direction. The figure shows the position of eight cells. The algorithm is based on the following assumptions:

1. Each flow has a position and altitude.
2. There is a β position around each flow. Each position has an altitude or objective function.
3. The amount of flow velocity is proportional to the slope.
4. The direction of current flow is from the cell with a higher altitude to the cell with a lower altitude.
5. The output point of the basin has the best value of the objective function and the optimal response.

First, the initial position of the current is defined:

$$flow(x_i) = lo + rand * (up - lo) \quad (10)$$

Where $flow(x_i)$: the location of the i^{th} flow, up : the upper bound of the decision variable, and lo : the lower bound of the decision variable

The following relation generates the position of each neighborhood:

$$neighbor(x_j) = flow(x_i) + randn * \Delta \quad (11)$$

Where $neighbor(x_j)$: location of j^{th} neighbor, $randn$: random value, and Δ : controller parameter.

If the Δ has large values, the algorithm searches for a large domain that leads to a global search, and if the Δ has small values, the algorithm searches for a small domain known as a local search. The following relationship is used to balance local and global search:

$$\Delta = (rand * x_{rand} - rand * flow(x_i)) * \| best(x) - flow(x_i) \| \omega \quad (12)$$

$$\omega = \left(\left(1 - \frac{iter}{iter_{max}} \right)^{2 * rand} \right) * \left(rand * \frac{iter}{iter_{max}} \right) \quad (13)$$

Where $iter_{max}$: maximum number of iterations, $iter$: current number of iteration, $best(x)$: the best location of flow, x_{rand} : random location that provided by Eq. 10, $rand$: random number.

The velocity of flow is updated through the following relation:

$$ve = rand * s_0 \quad (14)$$

Where s_0 : the slope vector between the neighbor and the current position of the flow and ve : the velocity of flow. The slope is computed as follows:

$$S_o(i, j, d) = \frac{flow(fitness(i)) - neighbor(fitness(j))}{\| flow(x_{i,d}) - neighbor(x_{j,d}) \|} \quad (15)$$

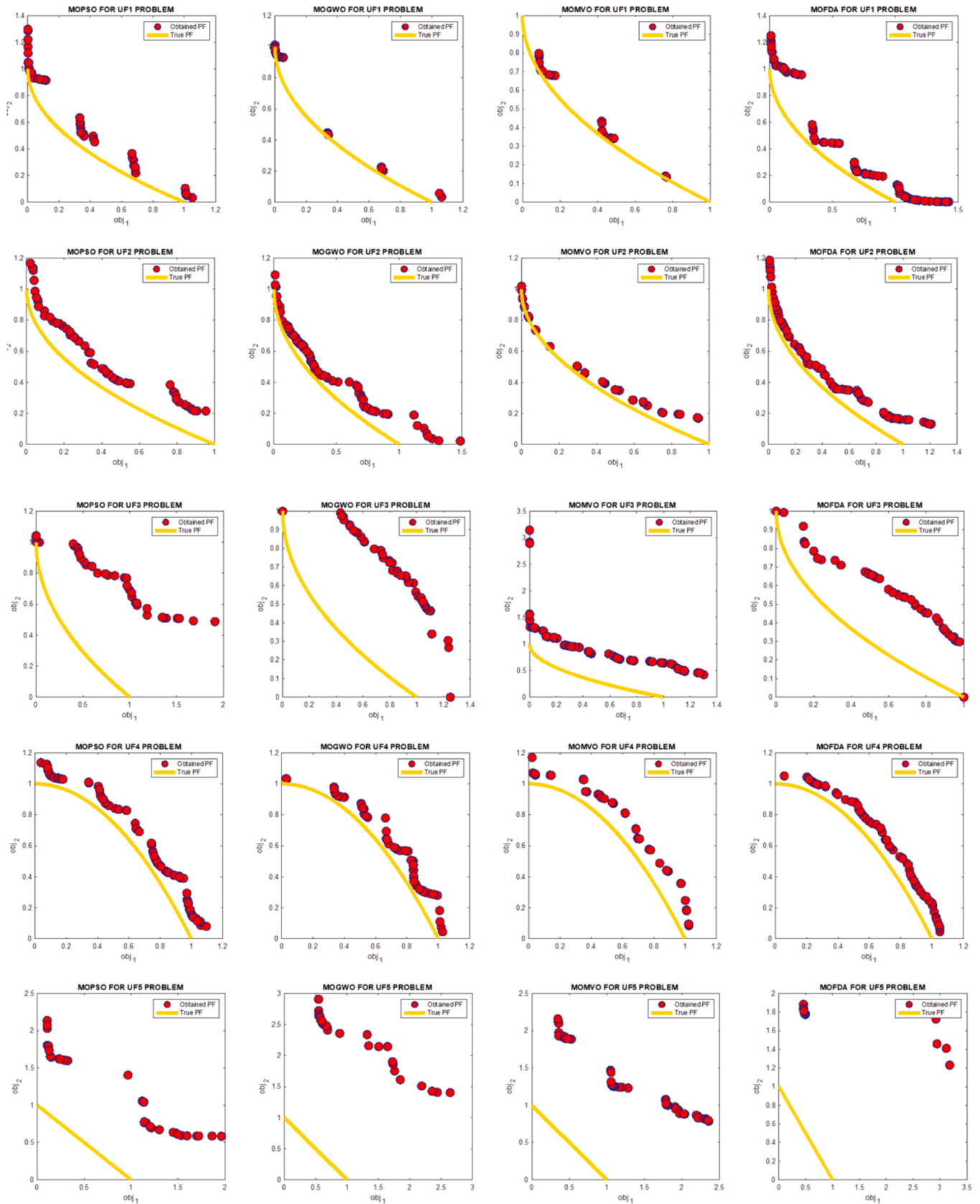


Fig. 4. Results obtained for UF1 to UF5.

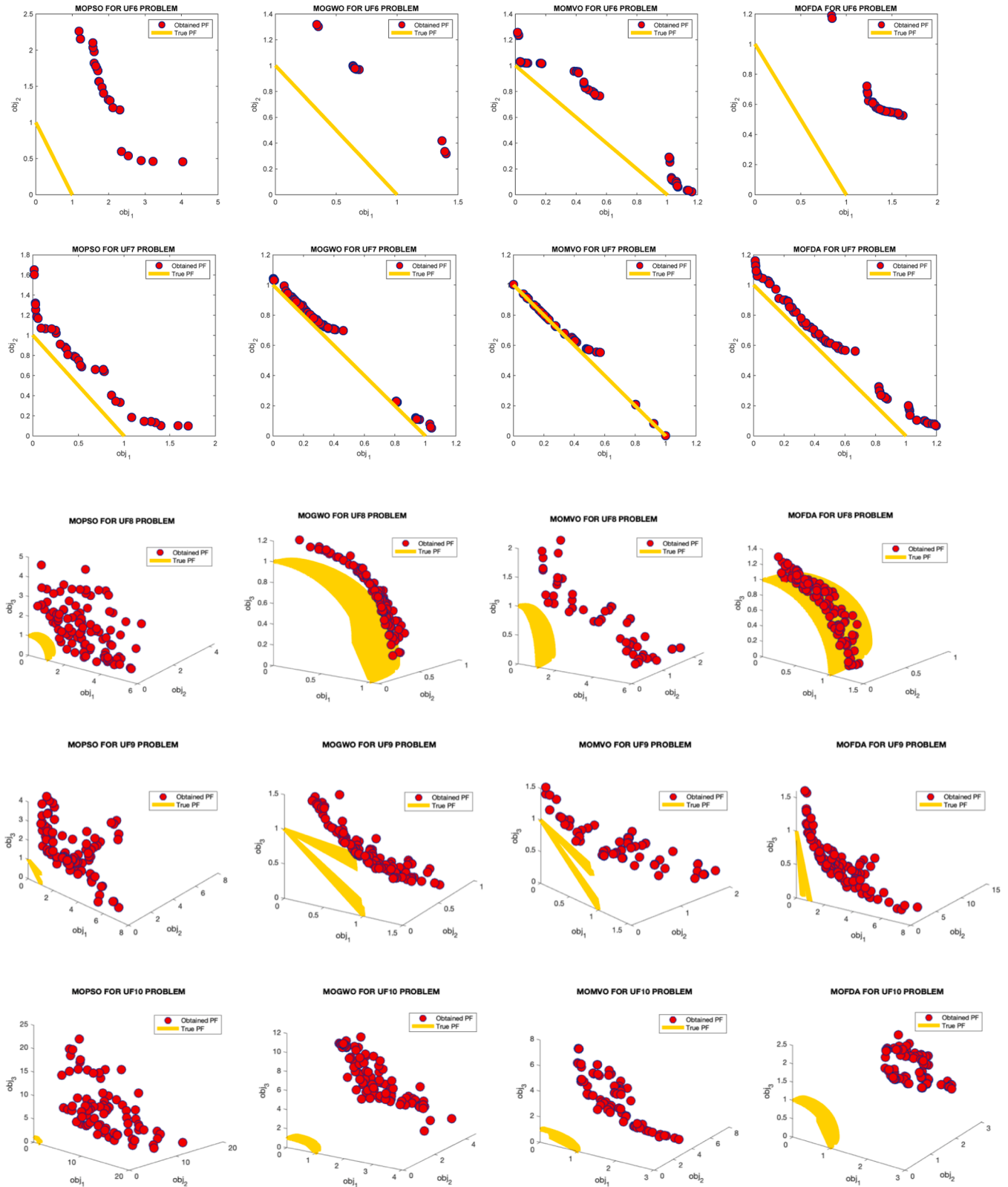


Fig. 5. Results obtained for UF6 to UF10.

Where $flow(fitness(i))$: the objective value for the flow i ,
 $neighbor(fitness(j))$: the objective value for the neighbor j .

The position of flow is computed as follows:

$$flow(new(x_i)) = flow(x_i) + v * \frac{flow(x_i) - neighbor(x_j)}{\|flow(x_i) - neighbor(x_j)\|} \quad (16)$$

Where $flow(new(x_i))$: the new location of the flow.

One of the FDA's capabilities is to fill sinks, which means getting out

Table 9

Comparison of algorithms using the GD performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
ZDT1	Ave	2.4377E-03	5.8093E-03	8.5601E-03	1.3995E-04
	SD	4.3009E-03	4.2240E-04	2.3452E-02	8.2243E-05
	WRST	+	+	+	
ZDT2	Ave	2.1019E-01	3.7956E-04	3.0091E-03	1.3455E-04
	SD	5.3241E-02	2.0061E-04	3.8970E-04	7.7779E-05
	WRST	+	+	+	
ZDT3	Ave	3.3490E-04	2.9332E-04	3.4539E-02	2.6161E-04
	SD	9.5468E-05	9.0168E-05	3.4350E-02	7.3852E-05
	WRST	+	+	+	
ZDT4	Ave	3.3419E+00	6.8746E+00	1.0091E+00	6.1853E-01
	SD	2.3209E+00	7.4640E+00	8.6578E-01	7.4213E-01
	WRST	+	+	+	
ZDT6	Ave	3.4466E-02	1.3278E-01	2.3240E-02	1.4840E-02
	SD	4.5460E-02	3.4317E-01	7.3432E-03	2.0896E-02
	WRST	+	+	+	
DTLZ2	Ave	9.5590E-03	1.2624E-03	4.4498E-03	3.5545E-02
	SD	7.6578E-04	5.2125E-04	6.4356E-04	1.3884E-02
	WRST	+	-	+	
DTLZ4	Ave	2.3245E-02	1.8707E-03	5.5467E-03	6.7278E-02
	SD	3.4350E-03	3.8729E-04	5.4435E-04	2.8453E-02
	WRST	+	-	+	
W^+ / W^-		14/0	10/4	14/0	
+/-/=		7/0/0	5/2/0	7/0/0	

Table 10

Comparison of algorithms using the IGD performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
ZDT1	Ave	4.4980E-04	1.2799E-03	1.6436E-03	3.6274E-04
	SD	1.3245E-03	3.4912E-04	1.4353E-04	4.2329E-05
	WRST	+	+	+	
ZDT2	Ave	2.2268E-02	2.3188E-02	1.8769E-03	1.4483E-03
	SD	8.9980E-03	8.6889E-03	5.3321E-03	5.0994E-03
	WRST	+	+	+	
ZDT3	Ave	2.4854E-04	4.8548E-04	1.5621E-03	2.3930E-04
	SD	3.8675E-05	6.8091E-05	2.5463E-04	3.1224E-05
	WRST	+	+	+	
ZDT4	Ave	1.2298E+00	2.6983E-01	2.3245E-01	1.3062E-01
	SD	3.0098E-01	3.3799E-01	1.2201E-01	1.6988E-01
	WRST	+	+	+	
ZDT6	Ave	6.0931E-04	7.5919E-04	5.1132E-04	4.8494E-04
	SD	2.3345E-02	8.5585E-03	1.8889E-04	1.0786E-04
	WRST	+	+	+	
DTLZ2	Ave	4.7980E-04	3.0120E-03	8.5998E-04	9.1326E-04
	SD	2.5908E-05	3.9605E-04	1.3908E-04	1.1362E-04
	WRST	-	+	+	
DTLZ4	Ave	1.6090E-03	7.4987E-03	3.9983E-03	3.4630E-03
	SD	9.3991E-05	3.7323E-04	8.4807E-04	8.5217E-04
	WRST	-	+	+	
W^+ / W^-		10/4	14/0	14/0	
+/-/=		5/2/0	7/0/0	7/0/0	

of the local optima. Suppose the value of a neighbor's objective function is not less than the current objective function in the present cell for a minimization problem. In that case, the sink filing process is applied to determine the direction of the current. At this point, the algorithm randomly selects a flow. If the objective function of the chosen flow is less than the objective function of the flow, the current will move in the same direction; otherwise, the flow will move in the direction of the dominant.

$$\left[\begin{array}{l} \text{If } (flow_fitness(r) < low_fitness(i)) \\ flow(new(x_i)) = flow(x_i) + rand * (flow(x_r) - flow(x_i)) \\ \text{else} \\ flow(new(x_i)) = flow(x_i) + 2 * rand * (best(x) - flow(x_i)) \end{array} \right] \quad (17)$$

The FDA algorithm is based on the following steps:

1. The initial matrix of the flow population is defined.
2. The value of the objective function is calculated for each member of the population matrix.
3. There are β neighborhoods with Δ neighborhood radius generated for each member of the population matrix.
4. The value of the objective function should be determined for each neighbor, and the best neighbor of each member should be determined according to the value of the objective function.
5. If the best neighborhood has a better objective function than the current flow, step 6 will be executed. Otherwise, we will go to step 7.
6. The flow velocity is obtained according to Eq. 14.
7. The position of the flow is updated based on Eq. 16.
8. The value of the objective function is calculated for the new position of the flow. If the objective part of the new position is better than the previous position, the new position of the flow is used instead of the previous position.
9. The stop rule should be controlled. If the stop rule is not satisfied, we return to step 3; otherwise, the optimization process is complete.

3.2. Multi-objective Flow Direction Algorithm

To propose the multi-objective variant of any meta-heuristic, the first step is to choose the type of multi-objective optimization. There are three algorithms based on the availability of decision-makers input on the importance (weight) of objective functions. If the significance of objectives is known before the optimization process, the algorithm is called a priori optimization algorithm. Algorithms in this category tend to aggregate the objectives into a single function using decision-maker weights. The algorithm is called interactive if decision-makers' inputs are considered during the optimization process. Such algorithms used "educated decisions" based on decision makers' preferences to "guide" the optimization process towards finding exciting solutions while saving on computational resources.

If decision makers' inputs are given a post-optimization process, the algorithm is called a posteriori optimization algorithm. The decision-making process is done after optimization, making it difficult for algorithm designers. With such critical information before the optimization process, we can find as many solutions as possible with enough diversity across all objectives to provide decision-makers with the best potential trade-offs. The three categories are shown in Fig. 2.

Due to the lack of decision-maker involvement in this work, we proposed a posteriori multi-objective version of the FDA called MOFDA. This algorithm can be used to find the best trade-offs between the objectives.

The single-objective version of the FDA was presented above. As was discussed, the best solution is considered as flow. The flow is used as a "flag" for promising regions of search space to search around it. In a single-objective problem, it is easy to use relational operators to find the best solution in each iteration. However, when solving a multi-objective problem posteriori, the best solution needs to be better defined. It depends on the decision makers' inputs, which is unknown in posterior multi-objective optimization. Therefore, we need to replace it with the best trade-offs of the solution obtained solution. Such solutions are called estimated Pareto optimal or non-dominated.

MOFDA was designed to solve single-objective optimization problems and cannot be used for multi-objective challenges. As a result, this study presents a multi-objective variant of MOFDA to tackle multi-criterion optimization issues. MOFDA contains three multi-objective optimization mechanisms similar to those employed by the multi-objective grey wolf optimizer (MOGWO) [33].

- Archive Mechanism:** It is a storage facility for storing or restoring the derived Pareto optimal solutions. The archive is updated in each iteration. It is limited, so solutions are discarded when full based on merit. The merit here is defined based on their distribution. If the solutions distribution is high in one region, it is

Table 11
Comparison of algorithms using the MS performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
ZDT1	Ave	9.9442E-01	8.2895E-01	8.9932E-01	9.9902E-01
	SD	3.4890E-02	7.4545E-02	7.5483E-01	2.1597E-03
	WRST	+	+	+	
ZDT2	Ave	6.3249E-02	6.3944E-01	8.9874E-01	9.4756E-01
	SD	3.3490E-01	3.4118E-01	3.7894E-02	2.2308E-02
	WRST	+	+	+	
ZDT3	Ave	9.8932E-01	9.7302E-01	1.3298E+00	9.9653E-01
	SD	2.4389E-03	2.1153E-02	4.8943E-01	3.4036E-03
	WRST	+	+	-	
ZDT4	Ave	3.4389E-01	6.2129E+00	3.3480E-01	2.0115E+00
	SD	1.1893E+00	6.9292E+00	9.3321E-02	2.0791E+00
	WRST	+	+	+	
ZDT6	Ave	1.9921E+00	2.1639E+00	1.2101E+00	1.4765E+00
	SD	3.3290E-01	1.6010E+00	1.9983E-01	6.6891E-01
	WRST	+	-	+	
DTLZ2	Ave	1.3219E+00	8.9087E-02	8.9932E-01	2.2604E+00
	SD	8.9289E-02	8.3330E-02	9.1003E-02	7.4746E-02
	WRST	+	+	+	
DTLZ4	Ave	1.3421E+00	2.1199E-01	9.9032E-01	2.8618E+00
	SD	2.8932E-01	6.5198E-02	8.8943E-02	4.0381E-02
	WRST	+	+	+	
W^+ / W^-		14/0	10/4	12/2	
+/-/=		7/0/0	5/2/0	6/1/0	

Table 12
Comparison of algorithms using the S performance metric.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
ZDT1	Ave	2.0021E-02	1.4894E-02	2.3290E-02	1.1232E-02
	SD	2.0322E-03	6.1897E-03	4.9323E-03	1.0403E-03
	WRST	+	+	+	
ZDT2	Ave	3.3202E-02	1.1600E-02	2.3210E-02	1.0840E-02
	SD	3.2311E-03	7.3141E-03	6.3466E-03	1.0079E-03
	WRST	+	+	+	
ZDT3	Ave	2.6548E-02	1.5123E-02	2.4536E-02	1.3132E-02
	SD	3.0061E-03	6.5176E-03	4.5099E-03	2.2505E-03
	WRST	+	+	+	
ZDT4	Ave	3.9087E-03	1.3446E+00	2.0985E-03	2.6242E-03
	SD	1.3852E-02	1.6767E+00	1.8168E-03	4.7661E-03
	WRST	+	+	-	
ZDT6	Ave	3.0009E-02	5.7668E-02	3.7886E-02	9.9020E-03
	SD	2.8976E-02	9.3879E-02	2.4536E-02	1.2236E-03
	WRST	+	+	+	
DTLZ2	Ave	6.4536E-02	4.7074E-03	8.7865E-03	7.2149E-02
	SD	5.8970E-03	1.2812E-03	6.9976E-03	1.5947E-02
	WRST	+	-	+	
DTLZ4	Ave	5.7865E-02	1.1737E-02	1.6574E-02	1.0427E-01
	SD	6.3425E-03	3.3199E-03	7.0098E-03	2.8690E-02
	WRST	+	-	+	
W^+ / W^-		14/0	10/4	12/2	
+/-/=		7/0/0	5/2/0	6/1/0	

better to remove some of them if the archive is full. The more crowded a solution, the more likely it is to be removed. The crowding distance will be used in this regard.

- II. **Grid mechanism:** It is an effective technique for enhancing non-dominated solutions in the archive. If the archive overflows, the grid technique will be used to rearrange the object space's division and discover the most occupied area, allowing one of the solutions to be eliminated. The additional member should be added to the least crowded segment to increase the diversity of the final approximated Pareto optimal front. The possibility of removing a solution grows as the number of possible solutions in the hypercube grows. If the archive is full, the busiest sections are chosen first, and a solution from one is removed randomly to make room for the new one. When a solution is placed outside the

hypercubes, a particular case arises. All segments in this scenario have been expanded to fit the most recent solutions. As a result, the segments of alternative solutions can also be changed.

- III. **Leader Selection Mechanism:** Solutions in a multi-objective search space are compared with this mechanism. To obtain a solution close to the global optimum, the search leaders direct the other search candidates to possible areas of the search space. As previously stated, the archive contains just the best non-dominated solutions, and the leader selection mechanism chooses the least crowded portions of the search space and presents the best non-dominated answers.

The selection for each hypercube is made using a roulette-wheel approach with the following probability:

$$P_i = \frac{C}{N_i} \quad (18)$$

where c is a constant number greater than one, and N is the variety of acquired Pareto optimal answers in the i th section.

Less crowded hypercubes have a more significant probability of suggesting new leaders, as shown in Eq. (18). When the number of obtained solutions in the hypercube is reduced, the probability of selecting a hypercube to select leaders from increases. We again need a merit factor when choosing Flow as the best solution, and we again need a merit factor. This is the opposite of updating the archive, as we want to drive the search process toward regions of the estimated Pareto optimal solutions that are less crowded. In this case, the less crowded around a solution, the more likely it is to be selected as the Flow. FDA, of course, provides the MOFDA algorithm with its convergence.

The MOFDA algorithm initiates by attempting to approximate the true Pareto optimal front for a multi-objective optimization challenge, beginning with a randomly selected set of solutions. Each solution undergoes evaluation against multiple objectives. Unlike single-objective optimization, where comparison between solutions is straightforward due to only one objective function, multi-objective optimization presents a complex scenario for comparison due to multiple criteria. In this context, a solution is considered to outperform (or dominate) another if it is equal to or better than the other in all objectives and superior in at least one. The algorithm maintains a record of all such non-dominated solutions in an archive. It iteratively updates the position of solutions

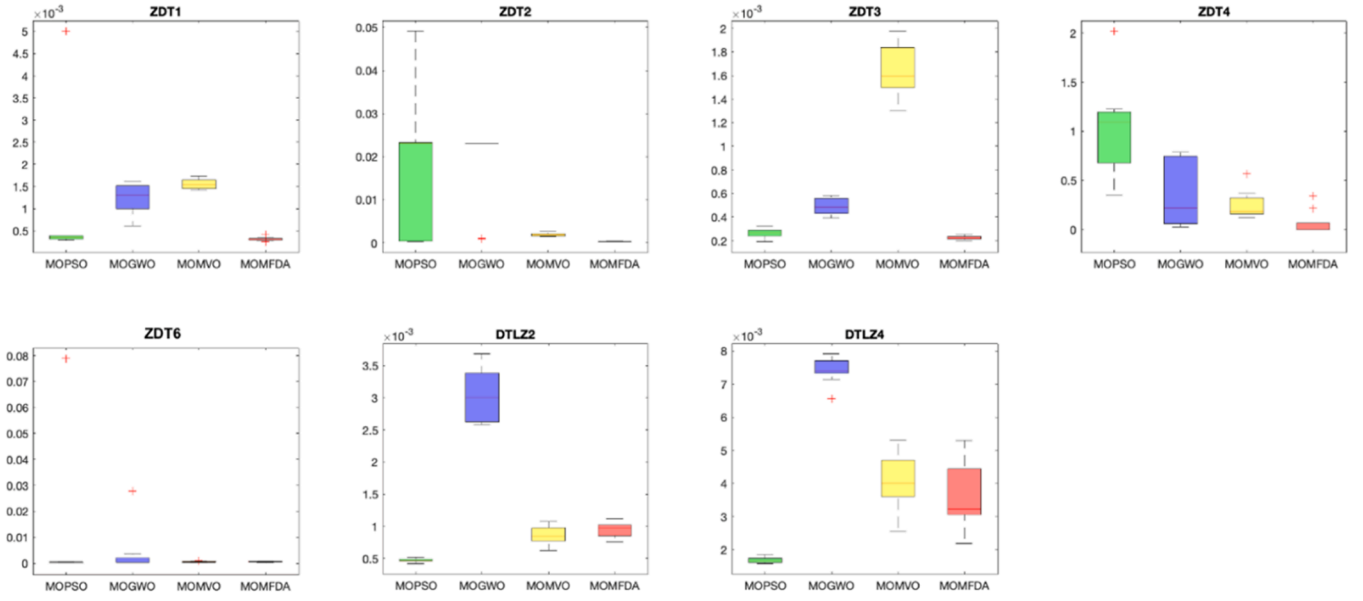


Fig. 6. Boxplot of the statistical results (IGD) for ZDT and DTLZ problems.

based on a specific equation, facilitating variable exchanges with solutions either in the archive or within the current non-dominated set. This dual approach helps in leveraging the best Pareto optimal solutions identified thus far and enhances the exploration of the search space. The refinement process persists until reaching a specific stopping point, with the algorithm also focusing on improving solution distribution across objectives by prioritizing less dense areas of the archive.

In multi-objective problems (MOPs) characterized by complex constraints, feasible regions often form intricate and constrained Pareto Fronts (PF) rather than true PFs. These challenges frequently result in algorithms being trapped in local optima, hindering their ability to achieve satisfactory convergence and distribution. While many MOP algorithms have been developed to address such issues, they often need help with narrow feasible spaces, multiple feasible zones, and intricate distributions. Balancing convergence, diversity, and feasibility remains a significant challenge for these methods.

Building on the foundation of the FDA algorithm, the proposed Multi-Objective FDA (MOFDA) adopts a similar strategy for enabling search agents to explore and exploit the search space effectively. MOFDA's unique strengths lie in its remarkable convergence and extensive coverage of the Pareto Front. The algorithm achieves exceptional convergence through its robust leader selection mechanism, ensuring that search agents are consistently guided toward optimal solutions. Additionally, its archive maintenance mechanism and its leader selection strategy drive extensive coverage. By systematically removing solutions from overcrowded segments and prioritizing leaders from sparsely populated regions of the archive, MOFDA significantly enhances solution diversity and achieves a well-spread Pareto Front across all objectives.

These innovations make MOFDA uniquely capable of addressing the limitations of existing MOP algorithms, particularly in scenarios with complex constraints and intricate feasible regions. The algorithm balances convergence, diversity, and feasibility and demonstrates superior performance in achieving a representative and evenly distributed set of solutions.

However, MOFDA distinguishes itself by utilizing an external archive to store non-dominated solutions and execute searches around these archived solutions. Nonetheless, MOFDA faces certain limitations:

- The algorithm is optimized for multi-objective problems with at most three or four objectives, beyond which its efficiency diminishes. This

decline in effectiveness with an increase in objectives is attributed to the rapid filling of the archive with non-dominated solutions, complicating the identification of optimal solutions for problems exceeding four objectives.

- MOFDA's utility is constrained and designed for problems with continuous variables. Thus, it is less suitable for discrete or mixed-integer optimization challenges.

4. Results

The results of MOFDA and other comparative algorithms are presented in this section. We first present the performance metric, experimental and truss structure benchmarks setup, followed by results analysis and discussion. MATLAB R2022a programmed the algorithms, and all datasets were evaluated in 20 independent runs on a computer with a CPU 2.3 GHz 8-Core Intel Core i9 with 16 GB of DDR4 RAM and MacOS BigSur as the operating system.

4.1. Performance metrics

In multi-objective optimization, researchers use multiple performance metrics to quantify the performance of a multi-objective algorithm from different perspectives. They can be categorized into coverage, convergence, and hybrid. Coverage metrics quantify the spread of estimated Pareto optimal solutions across all objectives. Convergence metrics quantify the accuracy of estimated Pareto optimal solutions. Hybrid metrics quantify and capture both coverage and convergence combined. In this paper, the following coverage and convergence metrics are used:

Generational Distance (GD): This represents the overall sum of the adjacent distances of solution candidates regarding different achieved sets using multiple algorithms, which is known as an intelligent indicator for evaluating the convergence characteristics of metaheuristic algorithms with multiple objectives [50].

$$GD = \left(\frac{1}{n_{pf}} \sum_{i=1}^{n_{pf}} dis_i^2 \right)^{\frac{1}{2}} \quad (19)$$

Spacing (S): A measure of the total distance between candidates regarding different achieved sets using multiple algorithms [51].

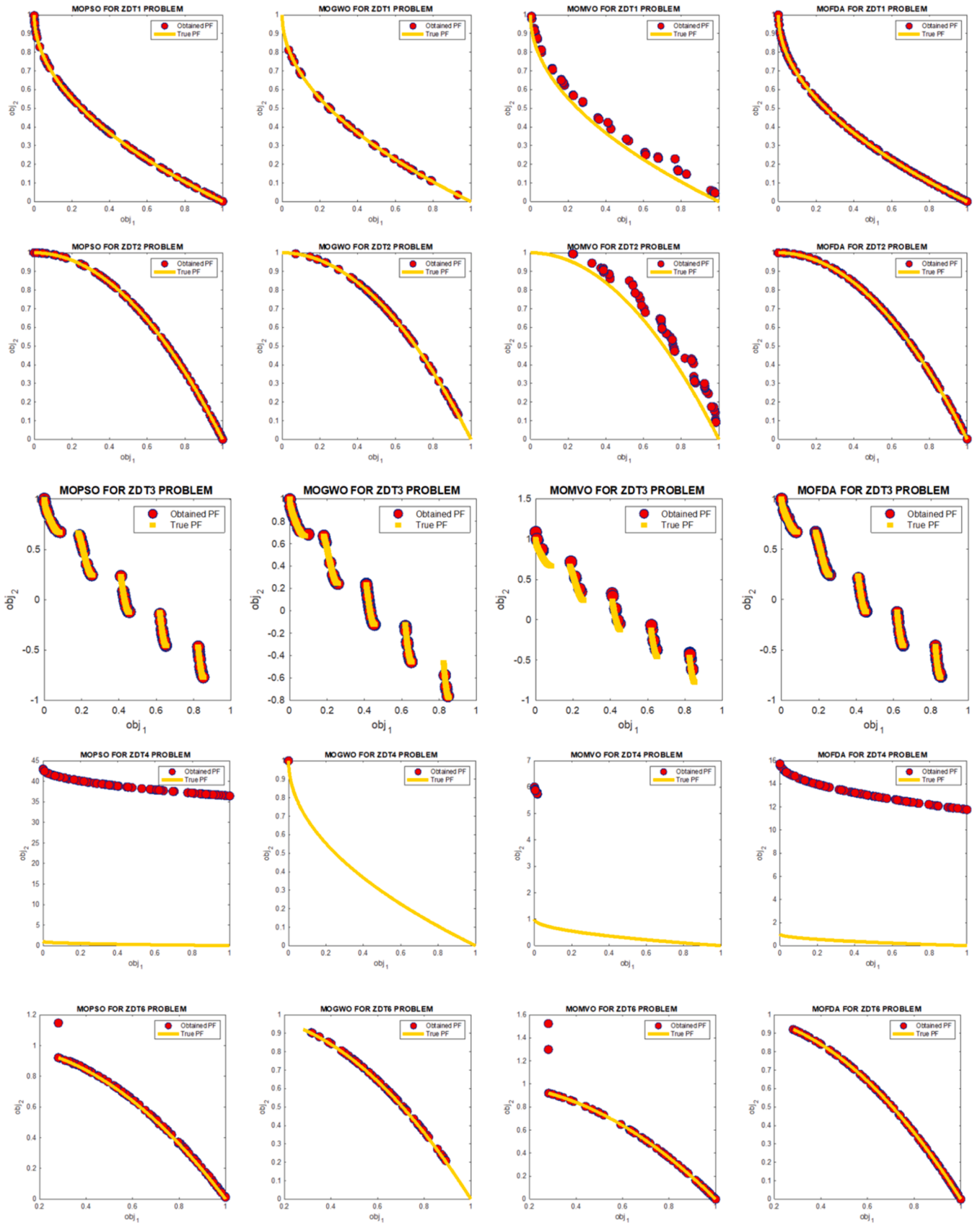


Fig. 7. Results obtained on ZDT problems

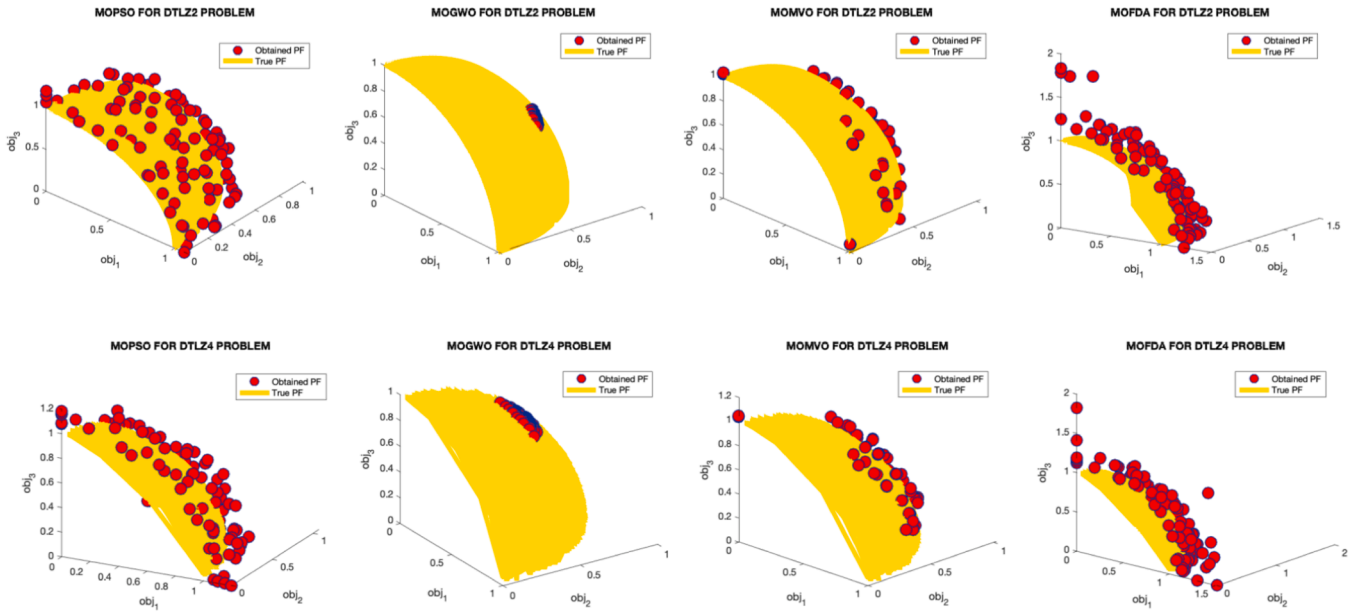


Fig. 8. Results obtained on DTLZ problems

Table 13
Comparison of algorithms using the GD performance metric on the engineering case studies.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
BNH	Ave	4.0094E-02	6.2465E-02	7.8974E-02	2.7807E-02
	SD	3.3242E-03	1.5645E-02	5.0976E-02	2.1220E-03
	WRST	+	+	+	
CONSTR	Ave	9.9898E-04	8.9220E-04	8.3420E-04	7.5776E-04
	SD	5.8976E-05	1.0565E-04	3.8976E-04	2.5866E-05
	WRST	+	+	+	
DISK BRAKEBREAKER	Ave	2.3240E-03	4.3311E-03	8.0097E-01	2.2549E-03
	SD	6.4532E-04	1.8074E-03	1.0876E-01	5.5667E-04
	WRST	+	+	+	
4-BAR TRUSS	Ave	1.2329E+01	1.1696E+01	1.0012E+01	1.5817E+01
	SD	9.8760E-01	2.3340E+00	3.6332E+00	1.3560E+00
	WRST	+	+	-	
WELDED BEAM	Ave	1.2134E-02	3.9207E-02	1.7843E-02	1.1751E-02
	SD	1.8769E-03	3.8097E-02	3.7432E-03	2.0775E-03
	WRST	+	+	+	
OSY	Ave	2.3492E+00	1.2736E+00	7.8769E-01	1.2404E+00
	SD	2.2341E+00	4.0179E-01	1.8973E-01	1.3041E-01
	WRST	+	+	-	
SPEED REDUCER	Ave	7.2576E+01	7.2876E+00	5.0013E+00	3.2095E+00
	SD	8.2316E+01	1.6129E+00	1.3241E+00	9.8580E-01
	WRST	+	+	+	
SRN	Ave	3.5821E-02	3.6743E-02	2.3003E-01	6.4322E-02
	SD	1.0032E-03	3.9924E-03	1.9843E-01	1.3210E-02
	WRST	-	+	+	
W^+/W^-		14/2	16/0	12/4	
$+/-/=$		7/1/0	8/0/0	6/2/0	

$$S = \left(\frac{1}{n_{pf}} \sum_{i=1}^{n_{pf}} (d_i - \bar{d})^2 \right)^{\frac{1}{2}} \quad \text{Where} \quad \bar{d} = \frac{1}{n_{pf}} \sum_{i=1}^{n_{pf}} d_i \quad (20)$$

Maximum Spread (MS): Represents the spread of candidates among other achieved sets by considering the distinct optimal choices [52].

$$MS = \left[\frac{1}{m} \sum_{i=1}^m \left[\frac{\min(f_i^{max}, F_i^{max}) - \max(f_i^{min}, F_i^{min})}{F_i^{max} - F_i^{min}} \right]^2 \right]^{\frac{1}{2}} \quad (21)$$

Inverted Generational Distance (IGD): A precise measure for the

performance estimation of the Pareto front approximations using the results of multiple many-objective optimization algorithms [37,53].

$$IGD = \frac{\sqrt{\sum_{i=1}^n d_i^2}}{n} \quad (22)$$

Hypervolume (HV): It measures the total volume of the objective space dominated by the Pareto front and bounded by a reference point, thereby reflecting both the diversity and the proximity of solutions to the true Pareto front. HV provides a comprehensive way to evaluate multi-objective algorithms, balancing coverage and convergence trade-offs [54].

Space to extend (STE): Spacing [55] evaluates the relative distances

Table 14

Comparison of algorithms using the IGD performance metric on the engineering case studies.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
BNH	Ave	9.6895E-03	3.3817E-03	3.0091E-03	1.0276E-03
	SD	1.6543E-04	1.1513E-03	2.7896E-04	1.4232E-04
	WRST	+	+	+	
CONSTR	Ave	5.0987E-04	6.4460E-04	1.7654E-03	5.4855E-04
	SD	5.4532E-05	1.7697E-04	3.6076E-04	5.9247E-05
	WRST	-	+	+	
DISK BRAKEBREAK	Ave	6.0830E-04	7.2091E-04	1.3343E-03	6.5781E-04
	SD	5.4356E-05	1.1206E-04	2.0987E-04	7.8833E-05
	WRST	-	+	+	
4-BAR TRUSS	Ave	2.0910E-02	2.1409E-02	2.0998E-02	2.0005E-02
	SD	3.7032E-05	1.5445E-04	4.5473E-04	2.9531E-05
	WRST	+	+	+	
WELDED BEAM	Ave	6.3001E-04	1.3441E-03	1.9975E-03	6.1310E-04
	SD	4.5463E-05	3.9377E-04	4.0094E-04	4.5389E-05
	WRST	+	+	+	
OSY	Ave	1.3643E-02	7.6220E-03	5.7008E-03	5.1824E-03
	SD	8.5945E-03	6.4006E-04	7.7656E-04	1.2429E-03
	WRST	+	+	+	
SPEED REDUCER	Ave	4.1985E-02	1.4243E-02	8.6899E-03	6.2340E-03
	SD	7.0980E-02	3.2032E-03	2.1149E-03	1.1771E-04
	WRST	+	+	+	
SRN	Ave	4.3146E-04	2.3823E-03	1.0234E-03	2.9558E-04
	SD	1.0056E-04	1.0614E-03	2.4626E-04	4.4231E-05
	WRST	+	+	+	
W^+ / W^-		12/4	16/0	16/0	
$+/-/=$		6/2/0	8/0/0	8/0/0	

Table 15

Comparison of algorithms using the MS performance metric on the engineering case studies.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
BNH	Ave	1.0000E+00	8.7156E-01	9.9876E-01	1.0000E+00
	SD	0.0000E+00	6.6957E-02	2.6582E-02	0.0000E+00
	WRST	=	+	+	
CONSTR	Ave	9.8323E-01	9.7790E-01	8.9943E-01	9.9956E-01
	SD	7.9894E-03	2.0113E-02	1.6543E-02	6.5953E-03
	WRST	+	+	+	
DISK BRAKEBREAK	Ave	8.7854E-01	9.9582E-01	1.0043E+00	9.9925E-01
	SD	2.3421E-03	1.2807E-02	2.3427E-01	6.0556E-03
	WRST	+	+	-	
4-BAR TRUSS	Ave	1.0034E+00	1.3787E+00	1.2123E+00	1.4879E+00
	SD	6.0093E-04	4.6120E-02	8.3452E-02	2.0517E-03
	WRST	+	+	+	
WELDED BEAM	Ave	1.0002E+00	1.1107E+00	1.0340E+00	1.1760E+00
	SD	5.8943E-02	1.0194E-01	8.4391E-02	5.1083E-02
	WRST	+	+	+	
OSY	Ave	3.2390E-01	6.6490E-01	6.1006E-01	7.7244E-01
	SD	2.2398E-01	2.5118E-02	2.8014E-02	8.3626E-02
	WRST	+	+	+	
SPEED REDUCER	Ave	3.8903E-01	7.6707E-01	7.5430E-01	8.6519E-01
	SD	2.9012E-02	2.4182E-02	2.3488E-02	1.5545E-02
	WRST	+	+	+	
SRN	Ave	9.3321E-01	7.0014E-01	9.1432E-01	9.7238E-01
	SD	4.5432E-02	8.0177E-02	3.0203E-02	1.4025E-02
	WRST	+	+	+	
W^+ / W^-		12/0	16/0	14/2	
$+/-/=$		6/0/1	8/0/0	7/1/0	

between consecutive solutions within the resulting non-dominated set. The STE metric is defined as the ratio of spacing to extent, calculated using the following formulas:

$$Spacing = \frac{1}{|P| - 1} \sum_{i=1}^{|P|} (d_i - \bar{d})^2 \quad (23)$$

$$Extend = \sum_{i=1}^M |f_i^{max} - f_i^{min}| \quad (24)$$

$$STE = \frac{Spacing}{Extend} \quad (25)$$

Wilcoxon Rank-Sum Test (WRST): The Wilcoxon rank-sum test, a non-parametric statistical technique, is employed to evaluate whether multiple datasets have been obtained from an identical distribution. The present analysis employs a significance level of 5%. The utilization of leverage enables a comprehensive evaluation of the algorithm's efficacy. The null hypothesis proposes that no significant difference exists in their performance in a scenario of identical mean metrics generated by two

Table 16
Comparison of algorithms using the S performance metric on the engineering case studies.

Functions		Algorithm			
		MOPSO	MOGWO	MOMVO	MOFDA
BNH	Ave	1.2312E+00	1.7477E+00	1.3245E+00	1.0286E+00
	SD	2.3425E-01	4.7352E-01	5.4532E-01	1.2542E-01
	+		+	+	
CONSTR	Ave	6.0741E-02	5.4894E-02	5.3426E-02	5.0019E-02
	SD	8.3469E-03	8.3095E-03	1.2439E-02	7.1749E-03
	+		+	+	
DISK BRAKEBREAK	Ave	2.0092E-01	1.3331E-01	3.4320E-01	1.1212E-01
	SD	1.2628E-02	1.9321E-02	2.9084E-01	1.8255E-02
	+		+	+	
4-BAR TRUSS	Ave	6.5464E+00	5.9045E+00	4.7243E+00	5.8150E+00
	SD	3.4352E-01	1.7462E+00	2.0083E+00	2.3294E-01
	+		+	-	
WELDED BEAM	Ave	3.0332E-01	4.4716E-01	3.2967E-01	2.0604E-01
	SD	2.3242E-02	2.1585E-01	1.3425E-01	4.0835E-02
	+		+	+	
OSY	Ave	1.0243E+00	1.6275E+00	1.5931E+00	1.4545E+00
	SD	1.3242E+00	4.7312E-01	4.5439E-01	2.0544E-01
	-		+	+	
SPEED REDUCER	Ave	4.5437E+01	2.0624E+01	2.6784E+01	2.0500E+01
	SD	5.5436E+00	3.1861E+00	5.4532E+00	3.0767E+00
	WRST	+	+	+	
SRN	Ave	2.9843E+00	2.6491E+00	3.0018E+00	2.0076E+00
	SD	5.5324E-01	1.2558E+00	1.5405E+00	2.6870E-01
	WRST	+	+	+	
W^+/W^-		14/2	16/0	14/2	
+/-/=		7/1/0	8/0/0	7/1/0	

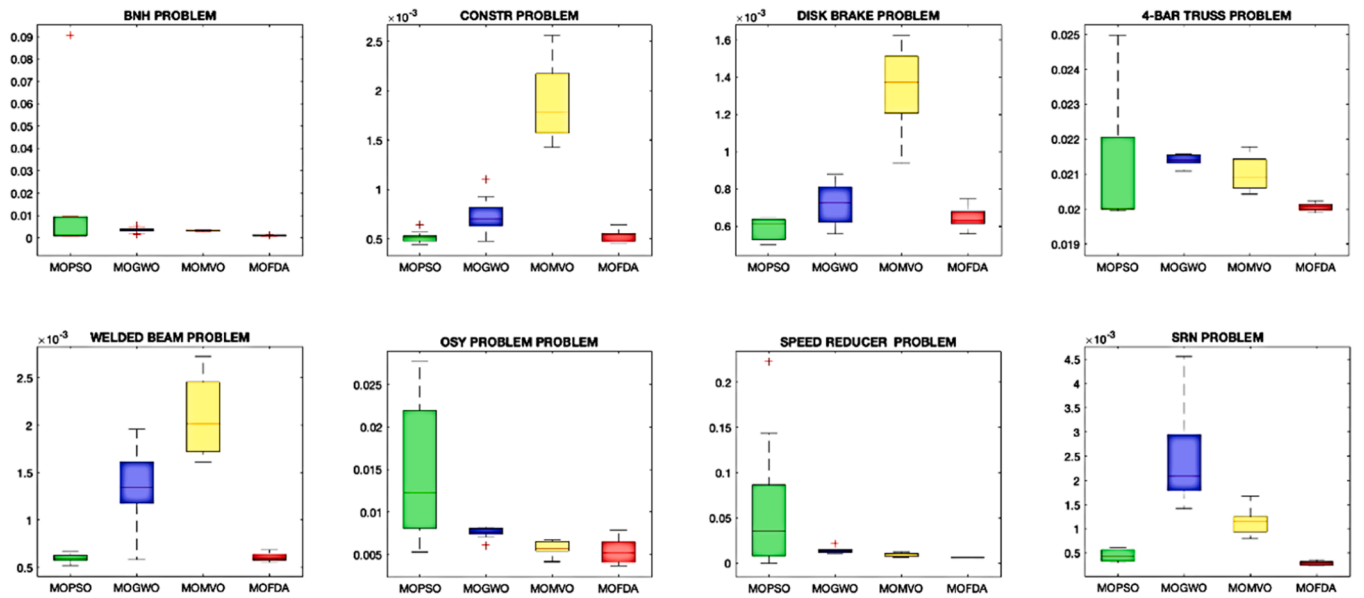


Fig. 9. Boxplot of the statistical results (IGD) for engineering problems.

compared algorithms. Conversely, the alternative hypothesis proposes a difference in the average performance measures of the examined algorithms. This research presents a comparative analysis of indicated algorithms, highlighting their differences through mathematical symbols such as subtraction, addition, and equality signs. These symbols denote the algorithm's performance as poor, significantly superior, or exhibiting no noticeable difference [53].

4.2. Experimental benchmarks setup comparing MOFDA with MOPSO, MOGWO, and MOMVO

The Multi-Objective Particle Swarm Optimization (MOPSO), Multi-Objective Grey Wolf Optimizer (MOGWO), and Multi-Objective Multi-

Verse Optimization (MOMVO) are compared to MOFDA in this section, and the best figure of a set of Pareto optimal is shown. Table 1 summarizes all of the listed algorithms' initial parameters. Notably, 100 populations and a maximum of 100,000 function evaluations were used for each experiment. To test the performance of the proposed algorithm, 25 different case studies:

- 17 two-objective and three-objective test functions (see Table 2,3, Table 4)
- 8 multi-objective simplified real-world case studies (see Appendix as supplementary material)

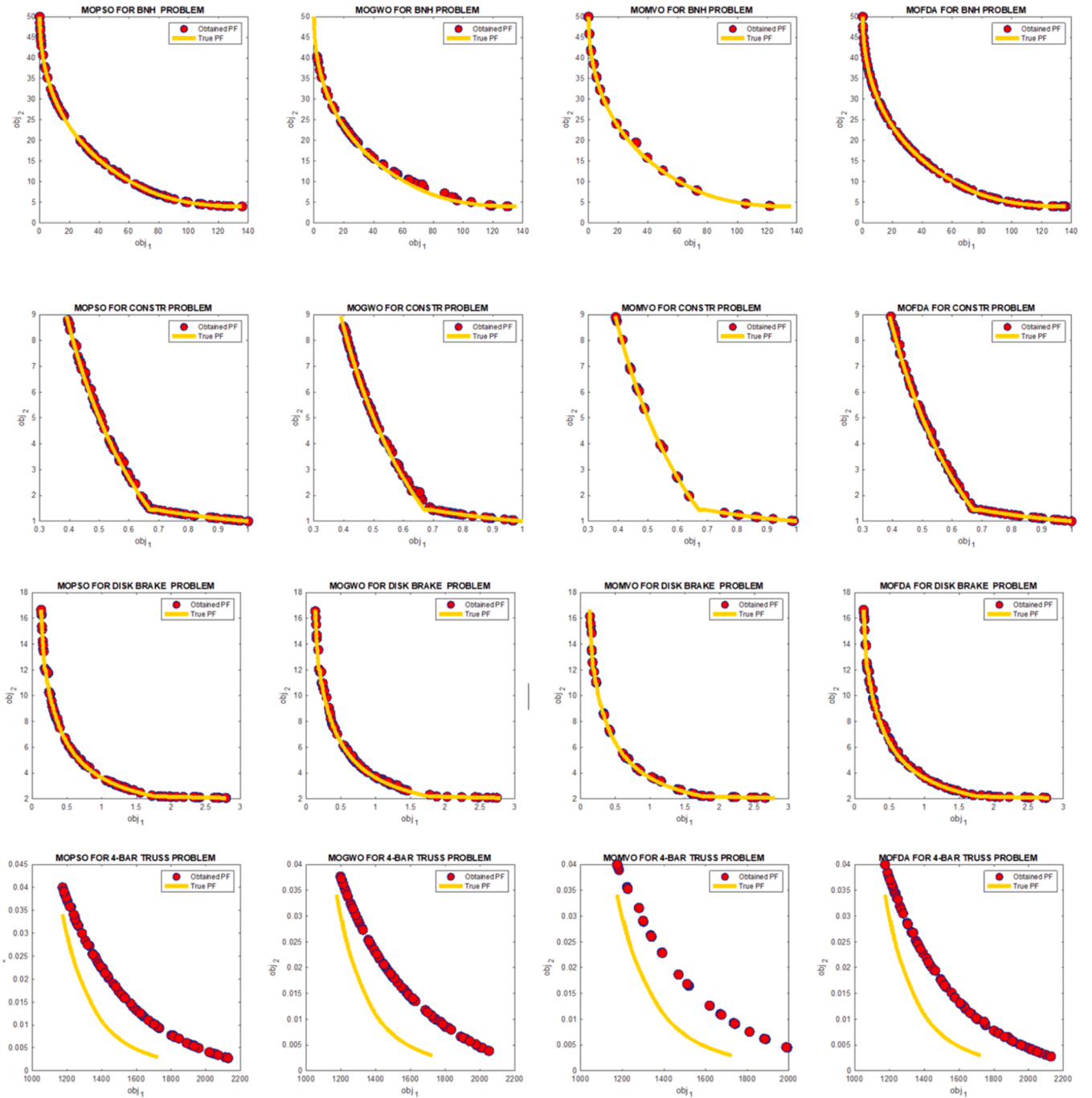


Fig. 10. Results obtained for BNH, CONSTR, DISK BRAKE, and 4-BAR TRUSS case studies

4.2.1. Discussion of the CEC-09 test function

In this section, the proposed algorithm is tested on CEC-09, and the obtained results are compared with other comparative methods in terms of IGD. It is clear from Table 5 that the proposed MOFDA got very comparative and better results in most of the tested cases. MOFDA got six of the best results out of ten test cases. MOMVO got the three best results out of ten test cases. MOGWO got the best result out of ten test cases. According to the standard deviation measure (SD), the results of the proposed MOFDA illustrated that it has a powerful ability to find almost the same results over different objectives. Table 6 shows the results of the proposed MOFDA compared to other methods in terms of GD. As illustrated in this table, it is clear that the proposed method got promising results compared to other advanced methods in this domain. MOFDA got five of the best results out of ten test cases. MOMVO got the

three best results out of ten test cases. MOGWO got the two best results out of ten test cases. According to the SD, the distribution of the obtained results proved the proposed method's ability to find the best results over many different runs. Moreover, MOFDA proved that it could see better results than other comparative methods associated with algorithm search methods.

From Table 7, the proposed MOFDA got promising and better results in most tested cases according to MS measure. MOFDA got seven of the best results out of ten test cases. MOPSO got the two best results out of ten test cases. MOMVO got the best result out of ten test cases. According to the standard deviation measure (SD), the results obtained by the proposed MOFDA proved that it has powerful search operators that can find almost the same promising results over different objectives. Table 8 shows the results of the proposed MOFDA compared to other methods in

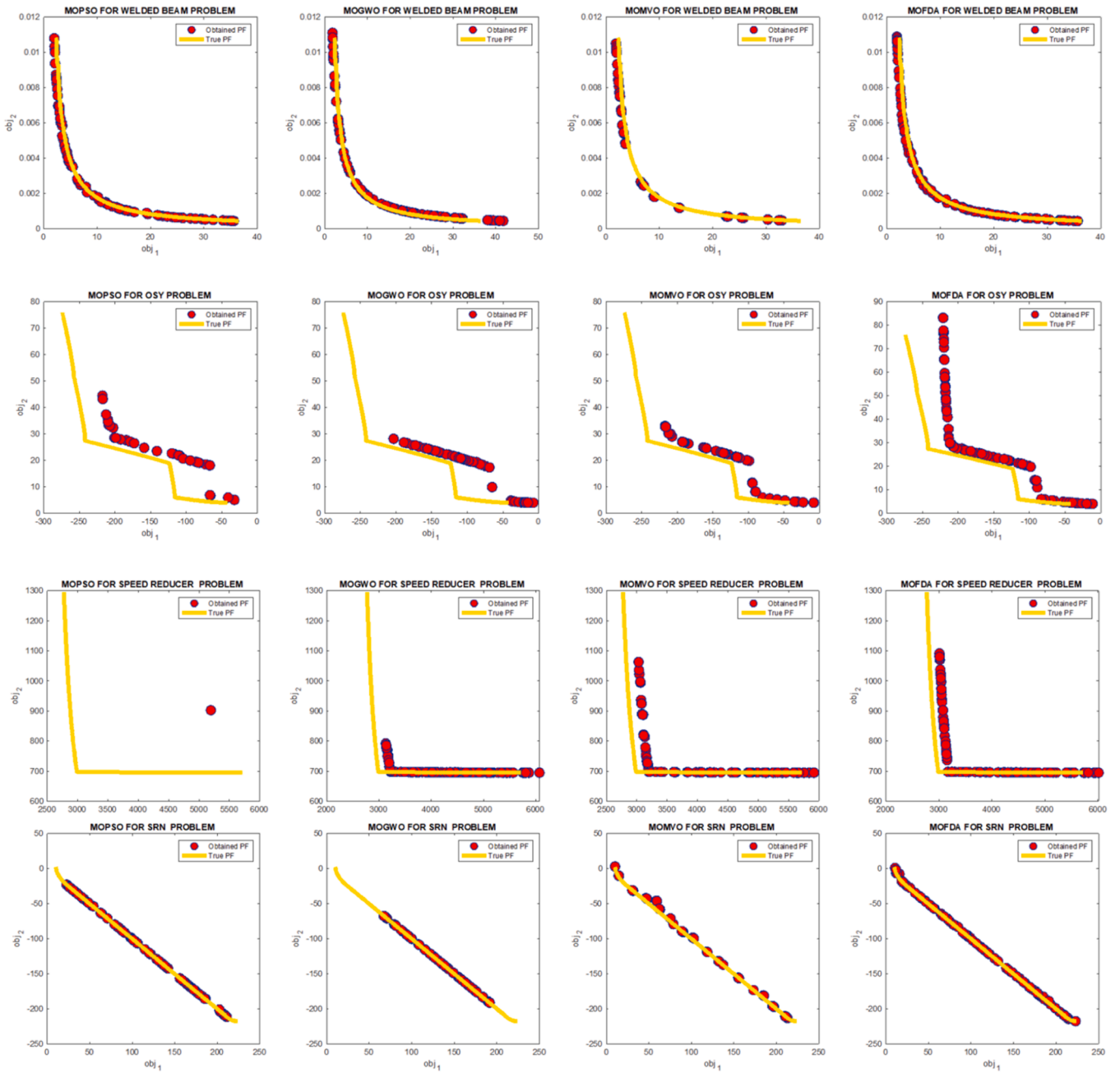


Fig. 11. Results obtained for WELDED BEAM, OSY, SPEED REDUCER, and SRN case studies

terms of S measure. It is evident that the proposed MOFDA got better results than other advanced multi-objective methods in this domain. MOFDA got six of the best results out of ten test cases. MOMVO got the four best results out of ten test cases. The results clearly illustrated the proposed method's ability to determine the best results over many different runs, according to the SD. Moreover, the procedure of the proposed MOFDA confirmed that the best results can always be achieved.

Fig. 3 shows boxplots of the statistical results (IGD) for all the tested methods on CEC-09 problems. When the distribution is low, the performance of the underlying algorithm is excellent; otherwise, the performance of the underlying algorithm is terrible. For example, in the third subfigure (UF3), the proposed MOFDA got very similar results over many different runs, and the other got some different results over the used runs. Another case (UF6) shows the proposed method's ability to find similar results compared to the other comparative methods.

Fig. 4 shows the true and obtained Pareto front for all the comparative methods on CEC-09 problems (UF1–5). For all the given cases, it is clear that the proposed method got very close optimal solutions to the given best solutions. The results are better than the other comparative methods for the first five cases. Fig. 5 shows the true and obtained Pareto front of all the comparative methods on CEC-09 problems (UF6–10). These figures also illustrate that the proposed MOFDA can find almost the best solutions for the given problems. The Wilcoxon rank sum test outcomes, depicted in Tables 5 through 8, convincingly show that MOGWO outperforms the other methods evaluated, and this superiority is significant. This reflects MOFDA's ability to deal with different challenging optimization problems.

This reflects MOFDA's ability to deal with different challenging optimization problems.

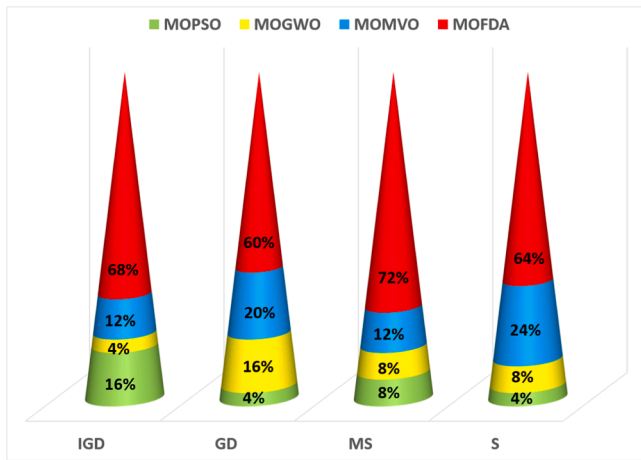


Fig. 12. Results obtained for 25 case studies

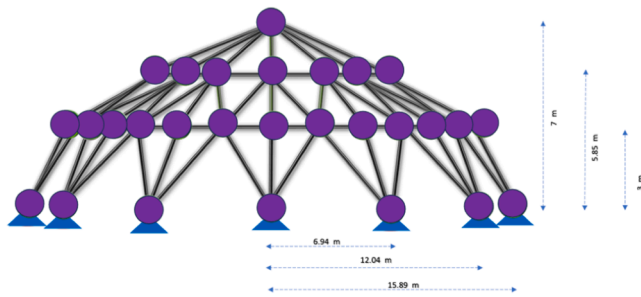


Fig. 13. 120-bar truss structure

4.2.2. Discussion of the ZDT and DTLZ test function

In this section, further assessments are performed to evaluate the performance of the proposed method (MOFDA) using advanced multi-objective benchmark problems (i.e., ZDT and DTLZ). This part supported the before-mentioned claims regarding the results obtained using the proposed method. In Table 9, the results obtained by the MOFDA are almost the best; it overcame the other comparative methods to solve this multi-objective problem. According to the GD measure, the best approach is MOFDA, followed by the MOGWO, which has the best results. The standard deviation of the given results is excellent, especially for the results achieved by the proposed MOFDA.

Table 10 shows the statistical results of the tested methods using ZDT and DTLZ regarding IGD measures. The proposed method is the best based on the best results achieved and the values of the distributions (SD). In the second place, MOPSO got some of the best remaining results; this indicates that MOPSO is promising at a specific level but different from the proposed MOFDA.

According to the obtained results in Table 11, the proposed method found excellent results in terms of MS measure. In most cases, it got better outcomes than the other comparative methods. It is clear that the structure of the proposed method helps in the search processes and facilitates the process of finding multi-objective solutions. Besides, the results of the proposed methods in terms of S performance measures are recorded in Table 12. According to this measure, the proposed method (MOFDA) also proved its ability to solve advanced multi-objective problems. When comparing the performance of the proposed method to the well-known methods, the proposed method got better results in most of the tested cases and, in the remaining, got comparative results. MOGWO got some best cases, illustrating its ability to be a comparator to the proposed MOFDA.

For further evaluation, the boxplot of the statistical results (IGD) for ZDT and DTLZ problems is presented in Fig. 6. These results illustrated

the diversity of the obtained results by evaluating all the obtained results using many runs. The proposed method got the best results using many different runs. The stability of the proposed method is more effective than the other comparative methods.

The Wilcoxon rank sum test outcomes, depicted in Tables 9 through 12, convincingly show that MOGND outperforms the other methods evaluated, and this superiority is significant. Figs. 7 and 8 show the true and obtained Pareto front for ZDT and DTLZ problems. These figures draw the original solutions and the best-obtained solution. In most issues tested problems, the proposed method got the most promising solution, which is very similar to the true solution for most tested cases.

4.2.3. Discussion of the multi-objective engineering design problems

This section tests the proposed MOFDA, which is tested to solve multi-objective industrial engineering design problems. These problems are BNH, CONSTR, DISK BREAK, 4-BAR TRUSS, WELDED BEAM, OSY, SPEED REDUCER, and SRN. Table 13 shows the results of the tested methods regarding GD measure. Clearly, the proposed method also has an excellent ability to solve multi-objective real engineering problems. This reflects the promising performance obtained by the proposed method for these cases. According to the standard deviation (SD), the variety of the obtained results is not high, which means that the exact solution for the same problem can be found (excellent stability).

The obtained results of the tested methods in terms of the IGD measure are given in Table 14. It is clear that after the best method, the proposed MOFDA, MOPSO got some of the best results in some cases. We can consider the proposed method as the best solution for solving real-world engineering problems, followed by the MOPSO. According to the MS measure, the results of the tested methods are given in Table 15, and the results in terms of the S measure are shown in Table 16. Based on these measures, the proposed method performed very well in almost all the tested cases except the CONSTR and DISK BREAK problems. MOPSO performed very well for these two problems. We concluded that the proposed method is the best competitor, followed by MOPSO and MOGWO.

Fig. 9 shows boxplots of the statistical results (IGD) for the tested engineering problems. The tested methods can find the exact solution in different runs. According to these criteria, the proposed method is the best for finding the solution over various runs. Fig. 10 shows the true and obtained Pareto front for engineering design problems (WELDED BEAM, OSY, SPEED REDUCER, and SRN). The results of the tested methods on various engineering design problems are presented. The findings of the Wilcoxon rank sum test, displayed in Tables 13 through 16, unequivocally indicate that MOGND outshines all other evaluated methods in terms of effectiveness. The difference between the proper solutions and the obtained ones is clearly shown in this figure. In most tested cases, the proposed method has promising solutions similar to the true ones. Fig. 11 shows the true and obtained Pareto front for engineering design problems (BNH, CONSTR, DISK BRAKE, and 4-BAR TRUSS). Also, the results that are given here encouraged our claim and proved the good performance of the proposed MOFDA in finding excellent solutions related to the true ones.

This figure clearly shows the difference between the true and obtained solutions. In most tested cases, the proposed method has promising solutions similar to the true ones. Fig. 10 shows the true and obtained Pareto front for engineering design problems (BNH, CONSTR, DISK BRAKE, and 4-BAR TRUSS). The results here encouraged our claim and proved the proposed MOFDA's good performance in finding excellent solutions related to the true ones.

The algorithms' performance in approximating Pareto optimal solutions is evaluated using the S and MS metrics, whereas their convergence is gauged with the IGD and GD performance metrics. To assess the algorithm's performance, a mean-based evaluation was executed, followed by a comparison using the Wilcoxon rank-sum test. According to the results obtained for all 25 mathematical benchmarks and real engineering designs, the algorithm demonstrated notable competitiveness

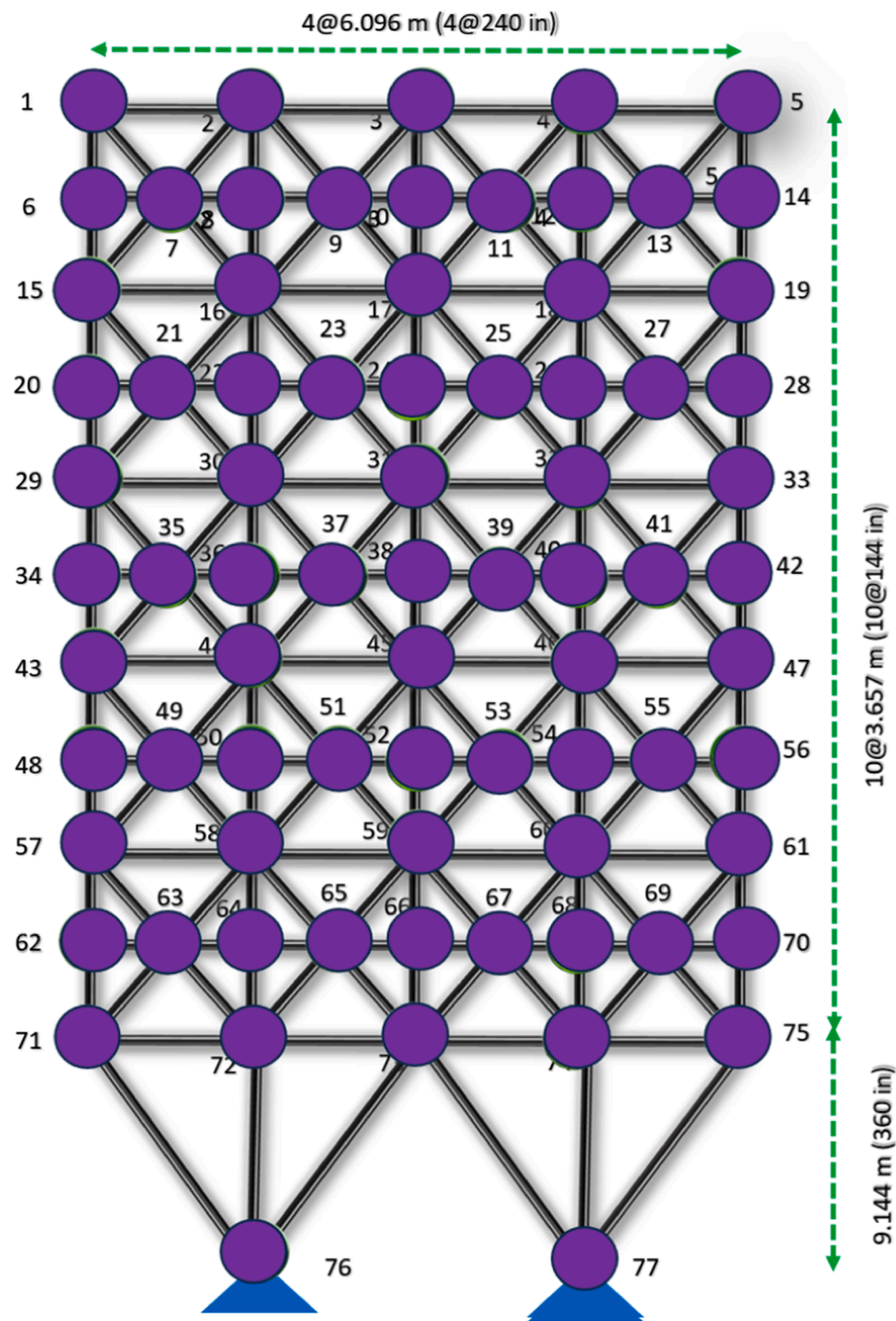


Fig. 14. 200-bar truss structure.

Table 17
120-bar truss loading condition.

Design variables	$X_i = 1, 2, \dots, 7$
Loading conditions (N)	$F_{213-36} = -5 \times 10^5$, $F_{237-48} = -15 \times 10^5$, $F_{249} = -3 \times 10^6$
Member grouping	$X_1(1 : 12)$, $X_2(13 : 24)$, $X_3(25 : 36)$, $X_4(37 : 60)$, $X_5(61 : 84)$, $X_6(85 : 96)$, $X_7(97 : 120)$

and effectiveness.

Fig. 12 shows four vertical graphs, each representing different performance metrics: IGD, GD, MS, and S. Within these cones, four methods are compared: MOPSO, MOGWO, MOMVO, and MOFDA. MOFDA demonstrates a predominant performance share, claiming 68%, 60%, 72%, and 64% of the total for IGD, GD, MS, and S, respectively. In comparison, MOMVO shows moderate percentages ranging from 12% to 24%, while MOPSO and MOGWO occupy the lower segments, fluctuating between 4% to 16% and 4% to 8% across the metrics. MOFDA remarkably outperforms other optimizers in terms of all metrics.

Table 18
200-bar truss loading condition.

Design variables	$X_i = 1, 2, \dots, 29$
Loading conditions (N)	$F_{y_i} = 10^4 \text{ i} = 1, 2, 3, 4, 5, 6, 8, 10, 12, 14, 15, 16, 17, 18, 19, 20, 22, 24, 26, 28, 29, 30, 31, 32, 33, 34, 36, 38, 40, 42, 43, 44, 45, 46, 47, 48, 50, 52, 54, 56, 57, 58, 59, 60, 61, 62, 64, 66, 68, 70, 71, 72, 73, 74, 75$ $F_{z_j} = -15 \times 10^5 \text{ j} = 1, 6, 15, 20, 29, 34, 43, 48, 57, 62, 71$
Member grouping	$X_1(1 : 4), X_2(5, 8, 11, 12, 17), X_3(19 : 24), X_4(18, 25, 56, 63, 94, 101, 132, 139, 170, 177)$ $X_5(26, 29, 32, 35, 38), X_6(6, 7, 9, 10, 12, 13, 15, 16, 27, 28, 30, 31, 33, 34, 36, 37),$ $X_7(39 : 42), X_8(43, 46, 49, 52, 55), X_9(57 : 62), X_{10}(64, 67, 70, 73, 76),$ $X_{11}(44, 45, 47, 48, 50, 51, 53, 54, 65, 66, 68, 71, 72, 74, 75),$ $X_{12}(77 : 80),$ $X_{13}(81, 84, 87, 90, 93), X_{14}(95 : 100), X_{15}(102, 105, 108, 111, 114),$ $X_{16}(82, 83, 85, 86, 88, 89, 91, 92, 103, 104, 106, 107, 109, 110, 112, 113), X_{17}(115 : 118)$ $X_{18}(119, 122, 125, 128, 131), X_{19}(135 : 138), X_{20}(140, 143, 146, 149, 153)$ $X_{21}(120 : 124, 126, 127, 129, 130, 141, 142, 144, 145, 147, 148, 150, 151), X_{22}(153 : 156)$ $X_{23}(157, 169, 163, 166, 169), X_{24}(171 : 176), X_{25}(178, 181, 184, 187, 190),$ $X_{26}(158, 159, 161, 162, 164, 165, 167, 168, 179, 180, 182, 183, 185, 186, 188, 189)$ $X_{27}(191 : 194), X_{28}(195, 197, 198, 200), X_{29}(196, 199)$

4.3. Truss Structure Benchmarks Setup Comparing MOFDA with MOHO, MOMVO2arc, and MOCO

This section evaluates two truss optimization issues, comparing Two-archive Multi-objective Multi-verse Optimizer (MOMVO2arc), Multi-objective Cheetah Optimizer (MOCO) and Multi-Objective Hippopotamus Optimizer (MOHO) with MOFDA. Initial parameters for all methods were obtained from their literature review. Notably, 100 populations and a maximum of 100,000 function evaluations were used for each experiment. Every test problem is converted to SI units and modified so that the problems are different from the original ones. These optimization challenges cover multi-functional truss sizing for 120-bar and 200-bar structures. The goals are to minimize structural mass and compliance within the allowed constraints of stress. Such problems are given in Eq. (26), where compliance is determined by Eq. (27).

$$\begin{aligned} \min (f_1, f_2) \\ \text{s.t. } \sigma_{\max} \leq \sigma_a \end{aligned} \quad (26)$$

$$\text{compliance} = \mathbf{u}^T \mathbf{F} \quad (27)$$

The displacement and loading vectors in these equations come from the finite element analysis where (f_1) and (f_2) are structural mass and compliance, respectively. Maximum stress ($\sigma_{\max}$) and allowable stress (σ_a) are the same for all problems. The material parameters, such as density (7850 kg/m³), elastic modulus (200 GPa), and allowable stress (400 MPa), are standardized. As structural components are typically discrete because of conventional beam dimensions, design variables are also discrete in this experiment. The schematic view for the 120-bar and 200-bar trusses is illustrated in Figs. 13 and 14. These included the 120-bar truss as a 3D space truss and the 200-bar truss as a 2D plane truss.

In some test cases, there are multiple design variables, so the design variables might not always be equal to the number of truss members. For the 120-bar and 200-bar problems, design variables are 7 and 29, respectively. Four performance measures, HV, IGD, and STE, are employed to evaluate the algorithms. HV assesses the distribution of the Pareto front. At the same time, GD and IGD measure the disparity between the obtained Pareto front and a reference front, representing either the true Pareto front or a non-dominated front derived from the solutions across all runs. The reference points for the test problems in this study are determined by the maximum objective function values obtained across all algorithms and optimization runs. Tables 17 and 18 provide detailed information on both examples' design variables, loading conditions, and member groupings.

The results in Table 19 demonstrate the superior performance of the MOFDA algorithm across all metrics for both the 120-bar and 200-bar truss optimization problems. For the 120-bar problem, MOFDA achieves the highest HV average of 8.4834E+10, with a standard deviation (SD) of 5.0741E+07, outperforming the next best algorithm, MOMVO2arc, which has an HV average of 8.4826E+10. In terms of IGD, MOFDA records the lowest average value of 7.2649E+02 with an SD of 3.3714E+02, indicating greater accuracy compared to MOMVO2arc (7.8749E+02), MOCO (1.4160E+03) and MOHO (2.4359E+03). For STE, MOFDA also achieves the best average value of 5.5198E-03 with an SD of 8.3967E-04, surpassing MOCO, MOHO, and MOMVO2arc.

Similarly, for the 200-bar problem, MOFDA maintains the highest HV average of 2.7375E+10 with the lowest SD of 2.4960E+07, slightly outperforming MOMVO2arc (HV: 2.7161E+10). The IGD metric for MOFDA is the lowest at 1.4317E+02 with an SD of 8.1239E+01,

Table 19
Comparison of algorithms using the HV, IGD, and STE performance metrics on the truss structures.

Truss Structure			Algorithm			
			MOHO	MOMVO2arc	MOCO	MOFDA
120-bar	HV	Ave	8.4367E+10	8.4826E+10	8.3834E+10	8.4834E+10
		SD	2.6169E+08	7.4391E+07	2.9996E+08	5.0741E+07
		WRST	+	+	+	
	IGD	Ave	2.4359E+03	7.8749E+02	1.4160E+03	7.2649E+02
		SD	1.0404E+03	3.5182E+02	9.2634E+02	3.3714E+02
		WRST	+	+	+	
	STE	Ave	6.4327E-03	5.7440E-03	8.8079E-03	5.5198E-03
		SD	9.4854E-04	1.0519E-03	3.3349E-03	8.3967E-04
		WRST	+	+	+	
200-bar	HV	Ave	2.6760E+10	2.7161E+10	2.5917E+10	2.7375E+10
		SD	3.0501E+08	1.7939E+08	2.0384E+08	2.4960E+07
		WRST	+	+	+	
	IGD	Ave	8.1144E+02	4.8163E+02	5.3741E+02	1.4317E+02
		SD	2.9663E+02	2.3598E+02	2.2775E+02	8.1239E+01
		WRST	+	+	+	
	STE	Ave	7.2357E-03	4.6357E-03	7.3142E-03	4.7558E-03
		SD	1.5058E-03	2.9100E-03	2.3126E-03	2.0662E-03
		WRST	+	-	+	
$\mathbf{W}^+ / \mathbf{W}^-$ +/-/=			12/0	10/2	12/0	
			6/0/0	5/1/0	6/0/0	

compared to others. For STE, MOMVO2arc achieves a strong average of 4.6357E-03 and MOHO with an SD of 1.5058E-03, outperforming MOFDA in AVE and SD. Based on the Wilcoxon rank-sum test results presented in Table 19 for both case studies, it is clear that the MOFDA algorithm outperforms other methods with a statistically significant efficiency level (see Table 19). MOFDA dominates the results with ten wins across both problems and demonstrates consistent performance across all metrics.

5. Conclusion

This study initially presents a literature review on multi-objective meta-heuristics. The multi-objective algorithms are classified based on the inclusion of decision-makers' input. A multi-objective variation of the FDA algorithm, featuring an archive, grid, and leader selection mechanism, is proposed. To substantiate the results, the proposed algorithm is contrasted with esteemed algorithms such as MOPSO, MOGWO, MOMVO, MOHO, MOCO, and MOMVO2arc. Performance evaluation of the multi-objective version of FDA is conducted on benchmark problems from the Completions on Evolutionary Computation (CEC-09) and on real-world engineering designs and truss structures coupled with some mathematical constraints (ZDT and DTLZ, for instance). The research reveals that the proposed MOFDA surpasses other methodologies in terms of IGD, GD, S, MS, HV, STE, and WRST indices and attains highly competitive outcomes. Comparing the true and achieved Pareto fronts allows the conclusion that the MOFDA can generate superior solutions closer to the Pareto front. The proposed algorithm is proven to significantly outshine other multi-objective algorithms in the literature.

The suggested MOFDA could be applied to future work, solving complex technical challenges and engineering design problems, such as truss structures and developing structural health assessments, which demand multi-modal and non-linear functions. Additionally, multiple comparative analyses with other renowned optimizers can be conducted to identify the most suitable optimizer for a specific design problem.

CRedit authorship contribution statement

Nima Khodadadi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Ehteram:** Conceptualization, Investigation, Data curation, Visualization. **Hojat Karami:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Mohammad H. Nadimi-Shahraki:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Laith Abualigah:** Formal analysis, Conceptualization. **Seyedali Mirjalili:** Funding acquisition, Formal analysis, Conceptualization, Investigation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Human participants or animals

This article does not contain studies with human participants or animals carried out by any of the authors.

Data availability

All data generated or analyzed during this study are included in this published article.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.rineng.2024.103670.

References

- [1] N. Khodadadi, E.-S.M. El-Kenawy, F. De Caso, A.H. Alharbi, D.S. Khafaga, A. Nanni, The Mountain Gazelle Optimizer for truss structures optimization, *Appl. Comput. Intell.* 3 (2023) 116–144.
- [2] B. Abdollahzadeh, N. Khodadadi, S. Barshandeh, P. Trojovský, F. S. Gharehchopogh, E.-S.M. El-kenawy, L. Abualigah, S. Mirjalili, Puma optimizer (PO): a novel metaheuristic optimization algorithm and its application in machine learning, *Cluster. Comput.* (2024) 1–49.
- [3] G. Dhiman, K.K. Singh, M. Soni, A. Nagar, M. Dehghani, A. Slowik, A. Kaur, A. Sharma, E.H. Houssein, K. Cengiz, MOSOA: a new multi-objective seagull optimization algorithm, *Expert. Syst. Appl.* 167 (2021) 114150.
- [4] A.M. Fathollahi-Fard, M. Hajiaghahi-Keshteli, R. Tavakkoli-Moghaddam, Red deer algorithm (RDA): a new nature-inspired meta-heuristic, *Soft. comput.* 24 (2020) 14637–14665.
- [5] S. Kaur, L.K. Awasthi, A.L. Sangal, G. Dhiman, Tunicate Swarm Algorithm: A new bio-inspired based metaheuristic paradigm for global optimization, *Eng. Appl. Artif. Intell.* 90 (2020) 103541.
- [6] E.-S.M. El-kenawy, N. Khodadadi, S. Mirjalili, A.A. Abdelhamid, M.M. Eid, A. Ibrahim, Greylag Goose optimization: nature-inspired optimization algorithm, *Expert. Syst. Appl.* 238 (2024) 122147.
- [7] H. Yapici, N. Cetinkaya, A new meta-heuristic optimizer: Pathfinder algorithm, *Appl. Soft. Comput.* 78 (2019) 545–568.
- [8] A. Kaveh, S. Talatahari, N. Khodadadi, Stochastic paint optimizer: theory and application in civil engineering, *Eng. Comput.* (2020) 1–32.
- [9] A. Mohammadi-Balani, M.D. Nayeri, A. Azar, M. Taghizadeh-Yazdi, Golden eagle optimizer: a nature-inspired metaheuristic algorithm, *Comput. Ind. Eng.* 152 (2021) 107050.
- [10] W. Zhao, L. Wang, Z. Zhang, H. Fan, J. Zhang, S. Mirjalili, N. Khodadadi, Q. Cao, Electric Eel Foraging Optimization: a new bio-inspired optimizer for engineering applications, *Expert. Syst. Appl.* 238 (2024) 122200.
- [11] I. Ahmadianfar, O. Bozorg-Haddad, X. Chu, Gradient-based optimizer: a new metaheuristic optimization algorithm, *Inf. Sci. (N Y)* 540 (2020) 131–159.
- [12] S. Arora, S. Singh, Butterfly optimization algorithm: a novel approach for global optimization, *Soft. comput.* 23 (2019) 715–734.
- [13] S. Alzoubi, L. Abualigah, M. Sharaf, M.S. Daoud, N. Khodadadi, H. Jia, Synergistic swarm optimization algorithm, (2024).
- [14] S. Shadravan, H.R. Naji, V.K. Bardsiri, The Sailfish Optimizer: a novel nature-inspired metaheuristic algorithm for solving constrained engineering optimization problems, *Eng. Appl. Artif. Intell.* 80 (2019) 20–34.
- [15] M.H. Amiri, N. Mehrabi Hashjin, M. Montazeri, S. Mirjalili, N. Khodadadi, Hippopotamus optimization algorithm: a novel nature-inspired optimization algorithm, *Sci. Rep.* 14 (2024) 5032.
- [16] Y. Lu, M. Liang, Z. Ye, L. Cao, Improved particle swarm optimization algorithm and its application in text feature selection, *Appl. Soft. Comput.* 35 (2015) 629–636.
- [17] N. Khodadadi, S.M. Mirjalili, S.Z. Mirjalili, S. Mirjalili, Chaotic Stochastic Paint Optimizer (CSPO), in: *Proceedings of 7th International Conference on Harmony Search, Soft Computing and Applications*, Springer, 2022, pp. 195–205.
- [18] M.H. Nadimi-Shahraki, S. Taghian, D. Javaheri, A.S. Sadiq, N. Khodadadi, S. Mirjalili, MTV-SCA: multi-trial vector-based sine cosine algorithm, *Cluster. Comput.* (2024) 1–45.
- [19] B. Zhang, X. Yang, B. Hu, Z. Liu, Z. Li, OEBBOA: A novel improved binary butterfly optimization approaches with various strategies for feature selection, *IEEE Access.* 8 (2020) 67799–67812.
- [20] N. Khodadadi, E. Khodadadi, Q. Al-Tashi, E.-S.M. El-Kenawy, L. Abualigah, S. J. Abdulkadir, A. Alqushaibi, S. Mirjalili, BAOA: Binary Arithmetic Optimization Algorithm With K-nearest Neighbor Classifier For Feature Selection, *IEEE Access.* 2023.
- [21] S. Arora, P. Anand, Binary butterfly optimization approaches for feature selection, *Expert. Syst. Appl.* 116 (2019) 147–160.
- [22] N. Khodadadi, E. Harati, F. De Caso, A. Nanni, Optimizing truss structures using composite materials under natural frequency constraints with a new hybrid algorithm based on cuckoo search and stochastic paint optimizer (CSSPO), *Buildings* 13 (2023) 1551.
- [23] A.N. Jadhav, N. Gomathi, WGC: Hybridization of exponential grey wolf optimizer with whale optimization for data clustering, *Alexandria Eng. J.* 57 (2018) 1569–1584.
- [24] F.S. Gharehchopogh, B. Abdollahzadeh, N. Khodadadi, S. Mirjalili, A hybrid african vulture optimization algorithm and harmony search: algorithm and application in clustering, *Adv. Swarm Intell.* (2023) 241–254.
- [25] L. Abualigah, A. Diabat, A comprehensive survey of the Grasshopper optimization algorithm: results, variants, and applications, *Neural Comput. Appl.* (2020) 1–24.
- [26] B. Abdollahzadeh, F.S. Gharehchopogh, N. Khodadadi, S. Mirjalili, Mountain Gazelle Optimizer: a new nature-inspired metaheuristic algorithm for global optimization problems, *Adv. Eng. Softw.* 174 (2022) 103282.
- [27] Q. Liu, X. Li, H. Liu, Z. Guo, Multi-objective metaheuristics for discrete optimization problems: a review of the state-of-the-art, *Appl. Soft. Comput.* 93 (2020) 106382.

- [28] N. Khodadadi, E. Khodadadi, B. Abdollahzadeh, E.-S.M. El-Kenawy, P. Mardanpour, W. Zhao, F.S. Gharehchopogh, S. Mirjalili, Multi-objective generalized normal distribution optimization: a novel algorithm for multi-objective problems, *Cluster. Comput.* (2024) 1–43.
- [29] N. Khodadadi, L. Abualigah, S. Mirjalili, Multi-objective Stochastic Paint Optimizer (MOSPO), *Neural Comput. Appl.* (2022), <https://doi.org/10.1007/s00521-022-07405-z>.
- [30] H. Karami, M.V. Anaraki, S. Farzin, S. Mirjalili, Flow Direction Algorithm (FDA): a novel optimization approach for solving optimization problems, *Comput. Ind. Eng.* 156 (2021) 107224.
- [31] D.H. Wolpert, W.G. Macready, No free lunch theorems for optimization, *IEEE Trans. Evol. Comput.* 1 (1997) 67–82.
- [32] C.A.C. Coello, M.S. Lechuga, MOPSO: a proposal for multiple objective particle swarm optimization, in: *Proceedings of the 2002 Congress on Evolutionary Computation. CEC'02 (Cat. No.02TH8600)* 2, 2002, pp. 1051–1056, <https://doi.org/10.1109/CEC.2002.1004388>, ppvol.
- [33] S. Mirjalili, S. Saremi, S.M. Mirjalili, L. dos S. Coelho, Multi-objective grey wolf optimizer: a novel algorithm for multi-criterion optimization, *Expert. Syst. Appl.* 47 (2016) 106–119.
- [34] S. Mirjalili, P. Jangir, S.Z. Mirjalili, S. Saremi, I.N. Trivedi, Optimization of problems with multiple objectives using the multi-verse optimization algorithm, *Knowl. Based. Syst.* 134 (2017) 50–71.
- [35] S. Kumar, N. Panagant, G.G. Tejani, N. Pholdee, S. Bureerat, N. Mashru, P. Patel, A two-archive multi-objective multi-verse optimizer for truss design, *Knowl. Based. Syst.* 270 (2023) 110529.
- [36] S. Kumar, G.G. Tejani, P. Mehta, S.M. Sait, A.R. Yildiz, S. Mirjalili, Optimization of truss structures using multi-objective cheetah optimizer, *Mech. Based Des. Struct. Mach.* (2024) 1–22.
- [37] N. Mashru, G.G. Tejani, P. Patel, M. Khishe, Optimal truss design with MOHO: a multi-objective optimization perspective, *PLoS. One* 19 (2024) e0308474.
- [38] S. Sharma, N. Khodadadi, A.K. Saha, F.S. Gharehchopogh, S. Mirjalili, Non-dominated sorting advanced butterfly optimization algorithm for multi-objective problems, *J. Bionic. Eng.* (2022) 1–25.
- [39] N. Khodadadi, L. Abualigah, E.-S.M. El-Kenawy, V. Snasel, S. Mirjalili, An Archive-based Multi-Objective Arithmetic Optimization Algorithm for Solving Industrial Engineering Problems, *IEEE Access*, 2022.
- [40] H.A. Abbass, R. Sarker, C. Newton, PDE: a Pareto-frontier differential evolution approach for multi-objective optimization problems, in: *Proceedings of the 2001 Congress on Evolutionary Computation (IEEE Cat. No.01TH8546)* 2, 2001, pp. 971–978, <https://doi.org/10.1109/CEC.2001.934295>, ppvol.
- [41] X. Li, A non-dominated sorting particle swarm optimizer for multiobjective optimization, in: *Genetic and Evolutionary Computation Conference*, Springer, 2003, pp. 37–48.
- [42] Q. Zhang, H. Li, MOEA/D: A Multiobjective Evolutionary Algorithm based on decomposition, *IEEE Trans. Evol. Comput.* 11 (2007) 712–731, <https://doi.org/10.1109/TEVC.2007.892759>.
- [43] K. Deb, S. Agrawal, A. Pratap, T. Meyarivan, A fast elitist non-dominated sorting genetic algorithm for multi-objective optimization: NSGA-II, in: *International Conference on Parallel Problem Solving from Nature*, Springer, 2000, pp. 849–858.
- [44] X.-S. Yang, S. Deb, Multiobjective cuckoo search for design optimization, *Comput. Oper. Res.* 40 (2013) 1616–1624.
- [45] F. Mendoza, J.L. Bernal-Agustin, J.A. Domínguez-Navarro, NSGA and SPEA applied to multiobjective design of power distribution systems, *IEEE Trans. Power Syst.* 21 (2006) 1938–1945.
- [46] A.R. Bhowmik, A.K. Chakraborty, Solution of optimal power flow using nondominated sorting multi objective gravitational search algorithm, *Int. J. Elect. Power Energy Syst.* 62 (2014) 323–334.
- [47] S. Mirjalili, P. Jangir, S. Saremi, Multi-objective ant lion optimizer: a multi-objective optimization algorithm for solving engineering problems, *Appl. Intell.* 46 (2017) 79–95.
- [48] S.Z. Mirjalili, S. Mirjalili, S. Saremi, H. Faris, I. Aljarah, Grasshopper optimization algorithm for multi-objective optimization problems, *Appl. Intell.* 48 (2018) 805–820.
- [49] N. Khodadadi, M. Azizi, S. Talatahari, P. Sareh, Multi-Objective Crystal Structure Algorithm (MOCryStAl): introduction and performance evaluation, *IEEE Access*. 9 (2021) 117795–117812.
- [50] D.A. Van Veldhuizen, G.B. Lamont, Multiobjective Evolutionary Algorithm research: A history and Analysis, Citeseer, 1998.
- [51] J.R. Schott, Fault tolerant design using single and multicriteria genetic algorithm optimization, (1995).
- [52] E. Zitzler, K. Deb, L. Thiele, Comparison of multiobjective evolutionary algorithms: empirical results, *Evol. Comput.* 8 (2000) 173–195.
- [53] B. Nouhi, N. Khodadadi, M. Azizi, S. Talatahari, A.H. Gandomi, Multi-Objective Material Generation Algorithm (MOMGA) for optimization purposes, *IEEE Access*. 10 (2022) 107095–107115.
- [54] A. Zhou, B.-Y. Qu, H. Li, S.-Z. Zhao, P.N. Suganthan, Q. Zhang, Multiobjective evolutionary algorithms: a survey of the state of the art, *Swarm. Evol. Comput.* 1 (2011) 32–49.
- [55] G.G. Tejani, N. Pholdee, S. Bureerat, D. Prayogo, A.H. Gandomi, Structural optimization using multi-objective modified adaptive symbiotic organisms search, *Expert. Syst. Appl.* 125 (2019) 425–441.